

Enabling brain-wide mapping of directed functional connectivity at 3T via layer-dependent fMRI with draining-vein suppression

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Wei-Tang Chang , Weili Lin, Kelly S. Giovanello

Biomedical Research Imaging Center, University of North Carolina at Chapel Hill, NC, USA • Department of Radiology, University of North Carolina at Chapel Hill, NC, USA • Department of Biomedical Engineering, University of North Carolina at Chapel Hill, NC, USA • Department of Psychology & Neuroscience, University of North Carolina at Chapel Hill, NC, USA

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Abstract

Layer-dependent functional magnetic resonance imaging (fMRI) offers a compelling avenue for investigating directed functional connectivity (FC). To construct a comprehensive map of brain-wide directed FC, several technical criteria must be met, including sub-mm spatial resolution, adequate temporal resolution, functional sensitivity, global brain coverage, and high spatial specificity. Although gradient echo (GE)-based echo planar imaging (EPI) is commonly used for rapid fMRI acquisition, it faces significant challenges due to the draining-vein effect, particularly when utilizing blood oxygen level-dependent (BOLD) contrast. In this study, we mitigated this effect by incorporating velocity-nulling (VN) gradients into a GE-BOLD fMRI sequence, opting for a 3T magnetic field strength over 7T. We also integrated several advanced techniques, such as simultaneous multi-slice (SMS) acceleration and NORDIC denoising, to enhance temporal resolution, spatial coverage, and signal sensitivity. Collectively, the VN fMRI method exhibited notable spatial specificity, as evidenced by the identification of double-peak activation patterns within the primary motor cortex (M1) during a finger-tapping task. Additionally, the technique demonstrated BOLD sensitivity in the lateral geniculate nucleus (LGN). Furthermore, our VN fMRI technique displayed superior robustness when compared to conventional fMRI approaches across participants. Our findings of directed FC elucidate several layer-specific functional relationships between different brain regions and align closely with existing literature. Given the widespread availability of 3T scanners, this technical advancement has the potential for significant impact across multiple domains of neuroscience research.

eLife assessment

This **useful** study presents a possible solution for a significant problem - that of draining vein sensitivity in functional MRI, which complicates the interpretability of laminar-fMRI results. The addition of a low diffusion-weighted gradient is presented to remove the draining vein signal and obtain functional responses with higher spatial fidelity. However, the strength of the evidence is **inadequate**, most tests appear to have been done only in a single subject. Significance thresholds in presented maps are very low and most cortical depth-dependent response profiles do not differ from baseline, even in the BOLD data shown as reference. Curiously, even BOLD group data fails to replicate the well-known pattern of draining towards the cortical surface.

Introduction

Layer-dependent functional magnetic resonance imaging (layer-dependent fMRI) is an emerging field which measures the layer-specific activity noninvasively in humans. The ability to extract layer-specific signal provides an exciting opportunity to dissociate between bottom-up feedforward (FF) and modulatory feedback (FB) responses which were activated in separated layers of a cortical unit. Among the fMRI methods, blood oxygen level dependent (BOLD) contrast has been the gold standard for nearly three decades (P. A. Bandettini et al. 1992 [↗](#); Ogawa et al. 1990 [↗](#)) due to its ability in rapid acquisition using Gradient Echo (GE)-based echo planar imaging (EPI) (P. Mansfield 1977 [↗](#)). However, the draining-vein effect poses a significant challenge to layer-dependent fMRI using BOLD contrast. This is primarily because BOLD contrast is T2*-based and sensitive to changes in magnetic field inhomogeneity caused by varying concentrations of deoxygenated hemoglobin. While arteries, capillaries, and draining veins all contribute to BOLD responses, the draining veins are considered the major contributor because of the larger volume and lower baseline oxygenation level compared to arteries (Boas et al. 2008 [↗](#)). The intracortical veins run perpendicular to the cortical surface and drain blood into pial veins. Consequently, BOLD responses induced in lower cortical layers will be carried to superficial layers, the so-called 'leakage model' (Markuerkiaga, Barth, and Norris 2016 [↗](#)).

To address the draining-vein issue in layer BOLD fMRI, one approach is to estimate depth-dependent BOLD signal by the inverse calculation of leakage model (Havlicek and Uludag 2020 [↗](#); Heinzle et al. 2016). However, the model-based vein removal methods are primarily exploratory and the effectiveness is still in need of validation (L. R. Huber et al. 2021 [↗](#)). Besides, the phase information of BOLD fMRI has also been used to reduce the draining-vein contamination. The BOLD signal arising from oriented vessels such as ascending and pial veins results in a coherent phase shift, while the BOLD signal from capillaries tends to have near-zero net phase (Menon 2002 [↗](#)). Therefore, the macrovascular signal variation may be captured by time-dependent phase variation and could be removed through phase regression. A recent study employed this phase regression technique to investigate laminar activity profiles and demonstrated that the phase-regressed results from GE-EPI are comparable to those from Spin Echo (SE) EPI, which is considered less susceptible to draining-vein contamination (Stanley et al. 2021 [↗](#)). Nevertheless, SE-EPI is not completely free from T2* effects either due to the long readout time (Goense and Logothetis 2008 [↗](#)) and, therefore, is not completely free from draining-vein effects. In another layer-dependent fMRI study focused on a finger-tapping task, phase regression significantly reduced the bias toward the superficial layer in the primary motor area (M1), although some residual macrovascular contribution remained (Knudsen et al. 2023 [↗](#)).

Apart from the BOLD contrast, vascular space occupancy (VASO) fMRI, a non-BOLD method, has been gaining popularity in recent years (L. Huber et al. 2021a [DOI](#)). VASO fMRI measures the changes of cerebral blood volume (CBV) rather than the level of blood oxygenation. These CBV changes primarily originate from small arterioles located close to layer-dependent neural activation sites (Drew, Shih, and Kleinfeld 2011 [DOI](#); Gagnon et al. 2015 [DOI](#)). In contrast, the larger vessels in the downstream are expected to have limited contribution on the overall CBV changes (Drew, Shih, and Kleinfeld 2011 [DOI](#); Gagnon et al. 2015 [DOI](#); O'Herron et al. 2016 [DOI](#); Takano et al. 2006 [DOI](#)). Although VASO imaging demonstrates high spatial specificity, the spatial coverage of layer-fMRI VASO acquisition is limited due to the nature of the contrast-generation mechanism (L. Huber et al. 2018 [DOI](#); L. Huber, Uludağ, and Möller 2019 [DOI](#)). Even with the advanced version of VASO, namely Multiple Acquisitions with Global Excitation Cycling (MAGEC) VASO, the acquisition time for a 83.2-mm-thick volume is 8.2 s (L. Huber et al. 2021b [DOI](#)). Additionally, it is challenging for VASO sequences to measure CBV in subcortical structures such as the lateral geniculate nucleus (LGN) because the arterial arrival time is too short for reproducible measurements (https://layerfmri.com/2021/02/22/vaso_ve/#more-2688 [DOI](#)). Even in the only VASO study that succeeded in measuring hippocampal CBV, the experiment required multiple inversion pulses, and the spatial resolution was relatively low ($1 \times 1 \times 4$ mm³) compared to that of layer-fMRI VASO (Rane et al. 2016 [DOI](#)).

In addition to layer-fMRI VASO method, several magnetization preparation sequences with turbo fast low angle shot (TurboFLASH) readout (Matthaei et al. 1985 [DOI](#)) have been proposed to improve spatial specificity. These include T1 ρ preparation, T2 preparation and diffusion-weighted T2 preparation. TurboFLASH readouts are used instead of EPI readouts to minimize T2* contamination. In the T1 ρ preparation sequence, the spin-lock editing pulses were used to enhance the contribution from microvasculature and reduce the influence of large veins (Rane et al. 2014 [DOI](#)). In the case of the T2 preparation sequence, the intravascular contribution to the BOLD signal can be reduced at high magnetic fields such as 7T (Hua et al. 2014). Nevertheless, the detection sensitivity and spatial specificity of T1 ρ and T2 preparation sequences were shown to be lower than a VASO sequence empirically (L. Huber, Hua, et al. 2017; L. Huber et al. 2021a [DOI](#)). To further enhance spatial specificity, small diffusion-weighted crusher gradients are included in the T2 preparation sequence. These small diffusion-weighted gradients, also referred to as velocity-nulling (VN) gradients, suppress signals from moving blood. Although the influence of large veins and arteries is reduced by the velocity-nulling gradients, the sensitivity of the diffusion-weighted T2 preparation sequence is likely compromised due to incomplete refocusing (L. Huber, Hua, et al. 2017).

To improve the detection sensitivity of sub-millimeter fMRI sequences with VN gradients, one possibility is to incorporate VN gradients into GE-EPI sequences. The sensitivity of GE-BOLD acquisition is 20-30% higher than SE-BOLD acquisition (Peter A. Bandettini et al. 1994 [DOI](#); Yacoub et al. 2005). However, the origins of the GE-BOLD signal are dominated by large vessels. By applying the VN gradients, the contribution from pial and draining veins could be substantially suppressed. Using GE-BOLD sequences also circumvents the issue of incomplete refocusing found in diffusion-weighted T2 preparation sequences. Nonetheless, the VN gradients, which were inserted between the RF pulse and the EPI readout, prolonged the echo time (TE), imposing the difficulty in maintaining the TE around the T2* value. This is especially problematic for sub-millimeter fMRI at 7T because the optimal TE is too short to accommodate the VN gradients.

This current study investigated the depth-dependent spatial specificity of sub-millimeter fMRI with VN gradients at 3T. The optimal echo time (TE) is in the range of 40-50 ms, matching the T2* value of grey matter at 3T, which is sufficiently long to incorporate VN gradients. To offset the reduction in the signal-to-noise ratio (SNR) when transitioning from 7T to 3T, the NOise Reduction with DIstribution Corrected principal component analysis (NORDIC PCA) was employed (Vizioli et al. 2021). The application of the NORDIC denoising method has been empirically shown to enhance the SNR by more than two-fold.

In this study, we initially implemented fMRI with a 0.9 mm isotropic resolution and VN gradients at 3T. We then examined the depth-dependent spatial specificity within the primary motor area (M1) using a finger-tapping experiment (L. Huber, Handwerker, et al. 2017). During the stage of optimizing the b value of the VN gradients, we restricted our acquisition to a slab encompassing M1 to avoid potential confounding from image reconstruction under a high acceleration rate. After determining a suitable b value, we employed the Simultaneous Multislice (SMS) method (Setsompop et al. 2012) for volumetric acquisition. Ultimately, we achieved the brain-wide acquisition with 0.9-mm isotropic resolution, 111.6 mm of field-of-view (FOV) thickness and repetition time (TR) of 3.84 s, thereby validating the depth-dependent spatial specificity. Utilizing a visual checkerboard experiment, we further demonstrated that the VN gradient led to only a marginal reduction in BOLD sensitivity at the lateral geniculate nucleus (LGN) – an area typically more challenging to measure compared to cortical regions. Building on the demonstrated BOLD sensitivity and spatial specificity, we explored the feasibility of conducting brain-wide directed functional connectivity studies at 3T. The results showcased several instances of directed functional connectivity that aligned with previous reports in the literature.

Theory

Draining-vein suppression using velocity-nulling gradient

The BOLD measurement at cortical layers is spatially blurred and biased towards superficial layers due to the draining-vein effect (Turner 2002 [↗](#); Duong et al. 2003 [↗](#)). Here we incorporate a velocity-nulling (VN) gradient into GE-EPI to suppress the signal from draining veins (see **Figure 1a** [↗](#)). The signal attenuation against the b value will generally follow the exponential decay in the Intravoxel Incoherent Motion (IVIM) process (Le Bihan 2019). The IVIM process models the collective motion of water molecules in blood within a vessel network as they transition from one vessel segment to another. This collective movement can be perceived as a pseudo-diffusion process where average displacements equate to the mean vessel segment length and the mean velocity matches that of the blood in the vessels.

In a capillary network, the average displacement is approximately 100 μm and the mean velocity is around 1 mm/s, yielding a pseudo-diffusion coefficient (D^*) of $10^{-8} \text{ m}^2/\text{s}$. In the case of cortical penetrating veins, the actual flow velocity is difficult to measure due to their small size. However, optical imaging suggests a mean flow velocity of $\sim 2.5 \text{ mm/s}$ in these veins (Shih et al. 2013 [↗](#)). Given the assumption that blood primarily flows along the penetrating veins, the average displacement is set as 3 mm, typically matching the cortical thickness of the human cortex. Accordingly, the pseudo-diffusion coefficient for penetrating veins is $\sim 7.5 \times 10^{-7} \text{ m}^2/\text{s}$. In contrast, the flow velocity in cortical arteries is $\sim 12 \text{ mm/s}$ (Schaffer et al. 2006), with an average displacement also of 3 mm. This configuration leads to a pseudo-diffusion coefficient of $\sim 3.6 \times 10^{-6} \text{ m}^2/\text{s}$.

The signal amplitude that underlies the effect of the VN gradient follows an exponential decay as represented by the equation $S/S_0 = \exp[-b(D^* + D_{\text{blood}})]$. In this formula, S denotes the vascular signal affected by the VN gradient, S_0 denotes the signal strength without the VN gradient, b denotes the b value of diffusion gradient, D^* indicates the pseudo diffusion coefficient, and D_{blood} indicates the water diffusion coefficient in blood. As the $D^* \gg D_{\text{blood}}$, the signal decay equation can be simplified to $S/S_0 = \exp(-b \cdot D^*)$. Using the predetermined pseudo diffusion coefficients for arteries, veins, and capillaries, we simulated the signal attenuations as illustrated in **Figure 1b** [↗](#). It is important to highlight that the b value represented in **Figure 1b** [↗](#) pertains to a single axis. Our results imply that a relatively small b value should effectively suppress the contribution from draining veins, yet simultaneously cause only minimal attenuation of signals originating from capillaries. Taking into account the varying orientation of penetrating veins within the highly convoluted human cortex, an equivalent degree of diffusion weighting is applied across all axes,

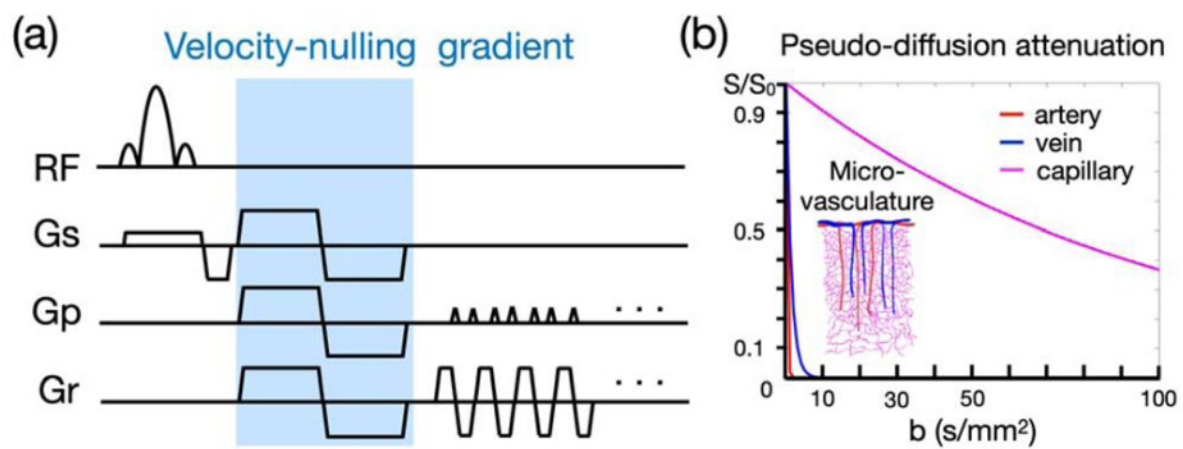


Figure 1

Velocity-nulling (VN) gradient in GE-EPI. (a) The diagram of pulse sequence. The VN gradient is highlighted in light blue. (b) The simulated signal attenuation against b values.

which triples the ultimate b value relative to that of a single axis. For example, using a b value of 30, equivalent to a single-axis b value of 10, can suppress the signal from veins to less than 0.1%, albeit with a consequential loss of 26% of the signal from capillaries.

Results

The efficacy of draining-vein suppression using velocity-nulling (VN) gradient

We evaluated the efficacy of draining-vein suppression by studying the depth-dependent profile of BOLD activation at M1, utilizing the finger-tapping task illustrated in [Figure 2a](#). We acquired slabs encompassing M1 with a range of b values of the VN gradient, as demonstrated in [Figure 2b](#). The left panel in [Figure 2c](#) presents the denoised slab image in three orthogonal views. The area around M1 was highlighted within a red box, and a zoom-in view of this region was shown in the upper-right panel. A color-coded layer representation within M1, overlaid on the T1 image, is shown in the bottom-right of [Figure 2c](#). The depth-dependent profiles of BOLD activation at M1 were extracted using the voxel-based approach. The depth-dependent profiles corresponding to a various of b values are shown in [Figure 2d-g](#). Overall, the results suggest that the VN gradient efficiently mitigated the draining-vein effect, thus achieving the desired laminar specificity at 3T. Both b values of 30 and 48 exhibited double-peak response patterns, yet the spatial variance in the profile with $b = 30$ was lower. Moreover, employing a b value of 48 may have overly suppressed the signal from capillaries, as simulated in [Figure 1b](#). Therefore, we utilized a b value of 30 for the VN gradient in the subsequent experiments.

The impact of SMS acceleration rate on depth-dependent profiles

To expand spatial coverage from a slab to a volume, both in-plane and SMS acceleration are required. The combined acceleration rate is defined as the product of in-plane and SMS accelerations. As higher acceleration rates may adversely affect SNR and signal quality, it is crucial to understand their impact on signal quality and the depth-dependent profile. Our tests on imaging protocols employed SMS factors of 4, 5, and 6 in combination with a VN gradient of $b = 30$. Additionally, we examined the protocol using an SMS factor of 4, but without a VN gradient. [Figure 3a](#)-[Figure 3d](#) showcase reconstructed images with SMS factors ranging from 4 to 6. The TRs corresponding to SMS factors of 4, 5, and 6 are 3.84s, 3.08s, and 2.58s, respectively, with an isotropic spatial resolution of 0.9 mm maintained throughout. The optimal image quality was realized with an SMS factor of 4. As the SMS factor was elevated from 5 to 6, minor image artifacts started to appear, particularly in the anterior subcortical and anterior frontal regions, as shown in [Figure 3c](#) and [Figure 3d](#). Similarly, the depth-dependent profiles of M1 activation in [Figure 3e](#) and [Figure 3f](#) display a double-peak pattern with an SMS factor of 4. This pattern demonstrated greater spatial stability with the application of the VN gradient, as indicated by increased statistical significance in the deeper layers (refer to [Figure 3f](#)). The depth-dependent profiles depicted in [Figure 3g](#) and [Figure 3h](#) showed increased spatial instability, likely attributable to feature loss when the combined acceleration rate was too high.

The impact of TE on depth-dependent profiles

To understand the impact of TE on the depth-dependent profiles within M1, a single participant completed four finger-tapping sessions with varied TEs. The resulting activation maps and depth-dependent profiles are displayed in [Figure 4](#). The TEs in conventional fMRI are 33, 38, and 43 ms are shown in [Figure 4a](#), [Figure 4b](#), and [Figure 4c](#), respectively, all without the application of the VN gradient. Those maps and profiles exhibited a notable bias towards the cerebral spinal fluid (CSF) located exterior to the cortical surface, with none displaying the double-

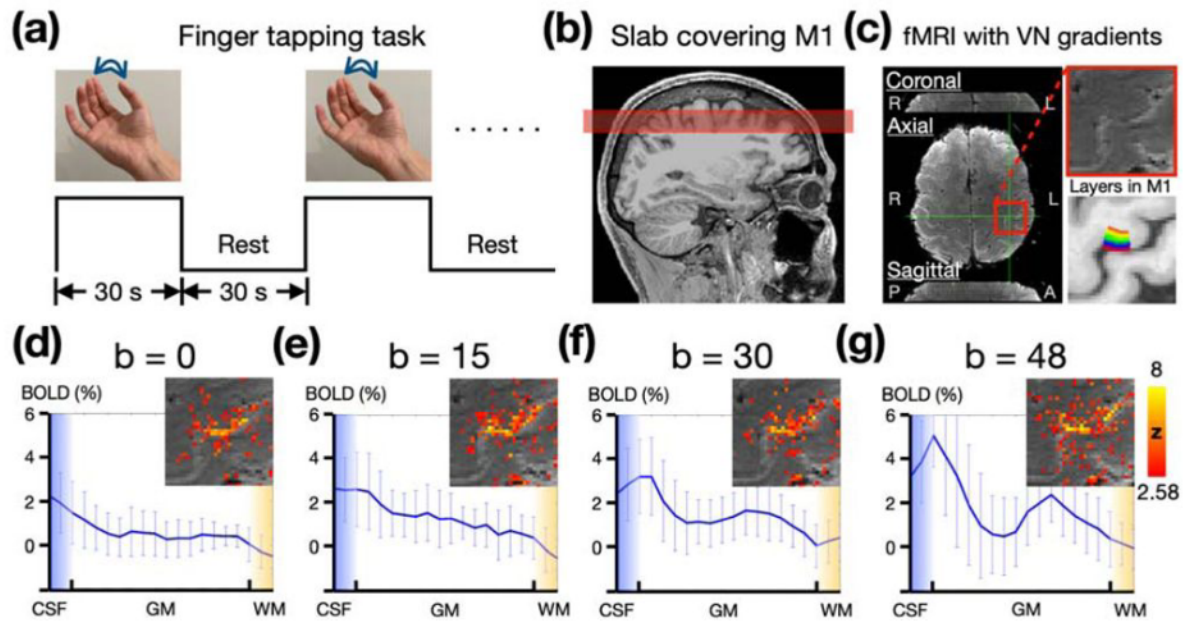


Figure 2

Empirical results of finger tapping task with different strength of draining-vein suppression from a single participant. (a) The paradigm of finger tapping task. (b) The illustration of the slab acquisition covering M1. (c) The slab image with 0.9-mm isotropic resolution in three orthogonal views. The 20 layers of M1 are color-coded as shown in the bottom-right panel. (d-g) The depth-dependent profiles of BOLD activation at M1 associated with $b=0, 15, 30, 48$. The statistical maps were corrected (uncorrected $p < 0.01$; corrected $p < 0.05$) and color-coded as indicated by the color bar.

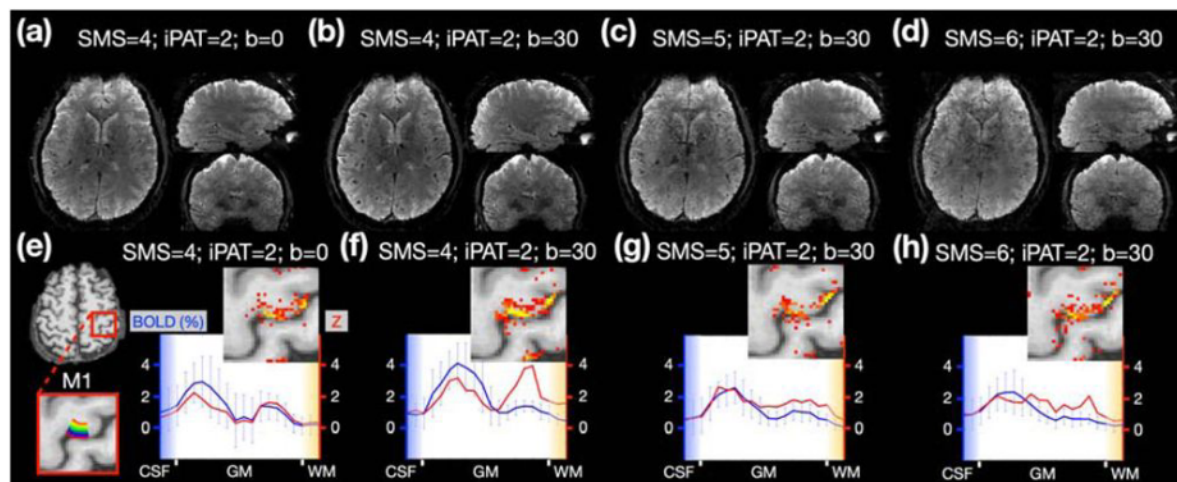


Figure 3

Sub-mm fMRI with brain-wide coverage in one participant. (a-d) The reconstructed images with different combination of SMS factor and b value. (e-h) The activation maps and depth-dependent profiles associated with different scanning protocols. The number of volumes acquired during the motor task was maintained at 282 across all protocols.

peak patterns. In contrast, the depth-dependent profile with the VN gradient, as shown in **Figure 4d**, displayed reduced bias towards CSF and exhibited the typical double-peak pattern. These results underscored the indispensable role of the VN gradient in draining-vein suppression.

The effectiveness of phase regression method for macrovascular suppression

To evaluate the impact of phase regression on BOLD activation maps as a function of TE, we applied phase regression to the fMRI datasets that were used in **Figure 4**, excluding the dataset with the VN gradient. In **Figure 5**, the left and middle columns represent BOLD activation maps without and with phase regression, respectively, while the right column displays the absolute difference between them. The rows correspond to TEs of 33 ms, 38 ms, and 43 ms, from top to bottom. The difference maps demonstrate that the impact of phase regression is most prominent at a TE of 33 ms, diminishing slightly as TE increases to 38 ms, and becoming minimal at 43 ms. Given that macrovascular contributions to the BOLD signal at 3T decrease with increasing TE (Irati Markuerkiaga et al. 2021), this reduced efficacy of phase regression along TE may be attributed to a diminished influence from large vessels. Moreover, at a TE of 43 ms, **Figure 5** indicates that phase regression has a marginal effect whereas **Figure 4** suggests that the VN gradient remains effective. This suggests that the impact of phase regression on smaller vessels, such as penetrating veins, may be limited.

BOLD sensitivity in subcortical regions

To analyze how BOLD sensitivity is influenced by the VN gradient, we applied a visual stimulus in the form of a checkerboard presentation, depicted in **Figure 6a**, to investigate the subcortical BOLD response in the LGN. As illustrated in the left panel of **Figure 6b**, the visual cortex elicited significant BOLD responses in the right hemisphere when the visual stimuli were introduced in the left visual hemifield. Similarly, stimuli in the right visual hemifield elicited BOLD responses in the left hemisphere. **Figure 6c** highlights the location of the LGN in green and displays the activation maps both without and with the VN gradient. To evaluate the impact of the VN gradient on the hemodynamic responses (HDRs) in the LGN, we averaged the corresponding event-related responses within the significantly-activated LGN voxels, and then performed an average across trials. The HDRs in the LGN, both without and with the VN gradient, are presented in **Figure 6d**. The time courses were re-sampled with a TR of 2s using linear interpolation for better visualization. Although the averaged HDR without the VN gradient demonstrated a slightly stronger BOLD response compared to the HDR with the VN gradient, this difference was marginal and exhibited no statistical significance. These findings suggest that the VN gradient does not significantly attenuate the activation strength within the LGN.

Robustness of depth-dependent activation profile

From **Figure 2** to **Figure 4**, the double-peak patterns were successfully demonstrated for a single participant. However, the consistency of this depth-dependent profile across a broader participant base has yet to be confirmed. Activation maps for six participants, both without and with the VN gradient, are shown in **Figure 7a** and **Figure 7b**, respectively. While one participant from the group without the VN gradient displayed the double-peak pattern, the majority of the depth-dependent profiles appeared spatially unstable. Consequently, the group-averaged depth-dependent profile failed to prominently exhibit the double-peak pattern. In contrast, in the VN fMRI data, the middle layers generally showed reduced activation. Hence, the double-peak pattern was evident in the group-averaged depth-dependent profile, as depicted in **Figure 7d**.

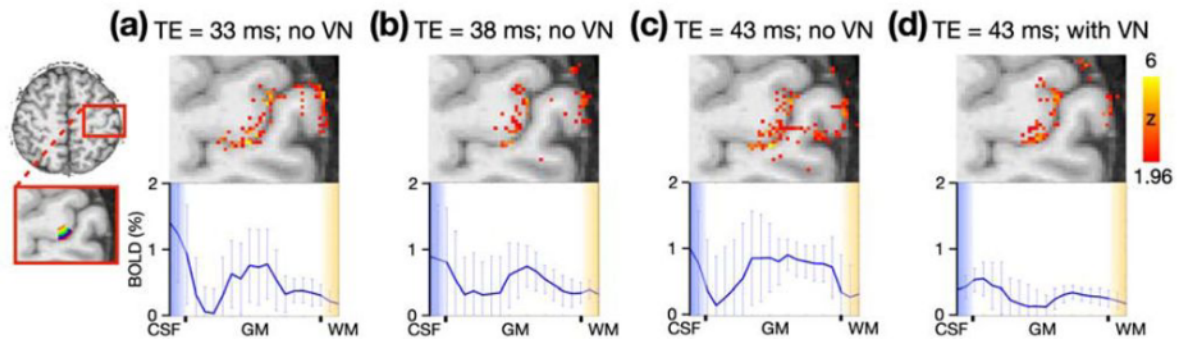


Figure 4

Illustration of TE impact on activation maps and depth-dependent profiles. A total of four finger-tapping experimental sessions were conducted on a single participant with varying TEs: (a) 33 ms, (b) 38 ms, (c) 43 ms, and (d) 43 ms with the VN gradient. No VN gradient was applied in (a)-(c). The red rectangle in the left-most column highlighted an enlarged view centered on the M1 region. Within M1, a total of 20 layers are represented and color-coded in the bottom-left panel. The statistical maps were corrected (uncorrected $p < 0.05$; corrected $p < 0.05$) and color-coded as indicated by the color bar on the far right. The images are presented in radiology view. The bottom row presents the enlarged views of the depth-dependent profiles, while the whiskers indicate the standard deviation across voxels within each individual layer.

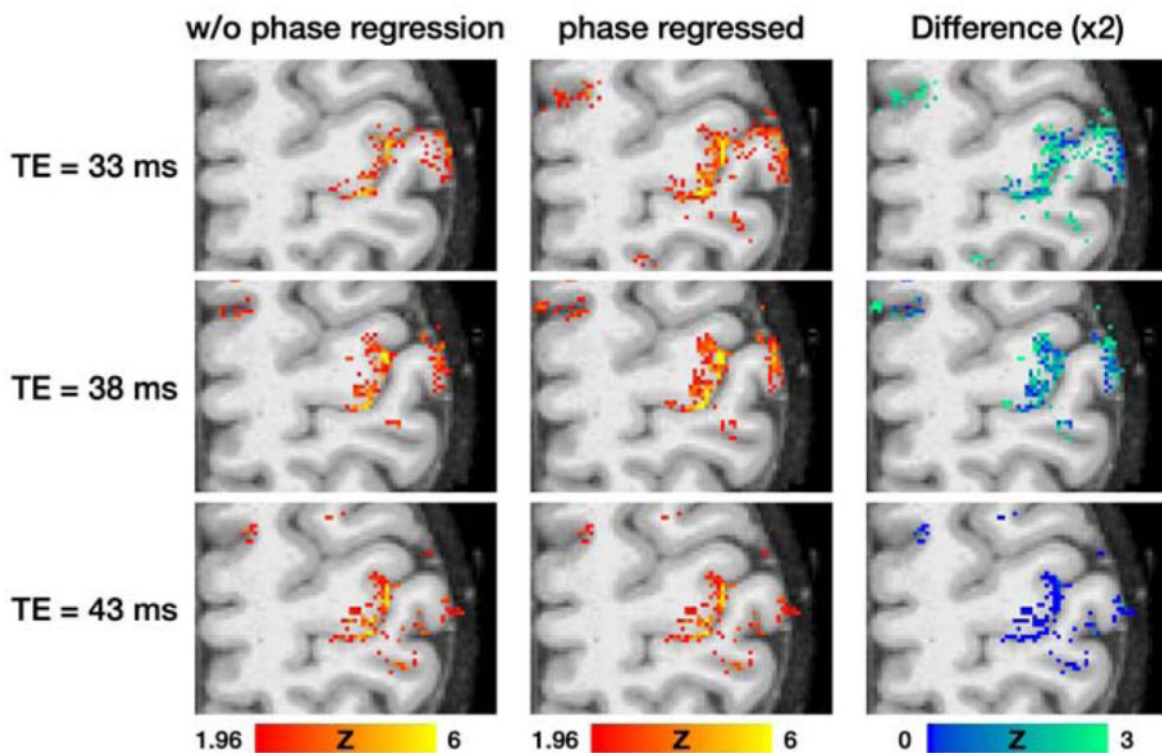


Figure 5

Influence of TE and phase regression on BOLD activation maps in a finger-tapping task. The rows represent the maps associated with TEs of 33 ms, 38 ms, and 43 ms, from top to bottom, respectively. Columns from left to right display the non-phase-regressed, phase-regressed, and difference maps, respectively. The right-most column depicts the absolute difference between the non-phase-regressed and phase-regressed maps. The activation maps were corrected statistically (uncorrected $p < 0.05$; corrected $p < 0.05$). The z values in each column are color-coded as indicated by the color bars.

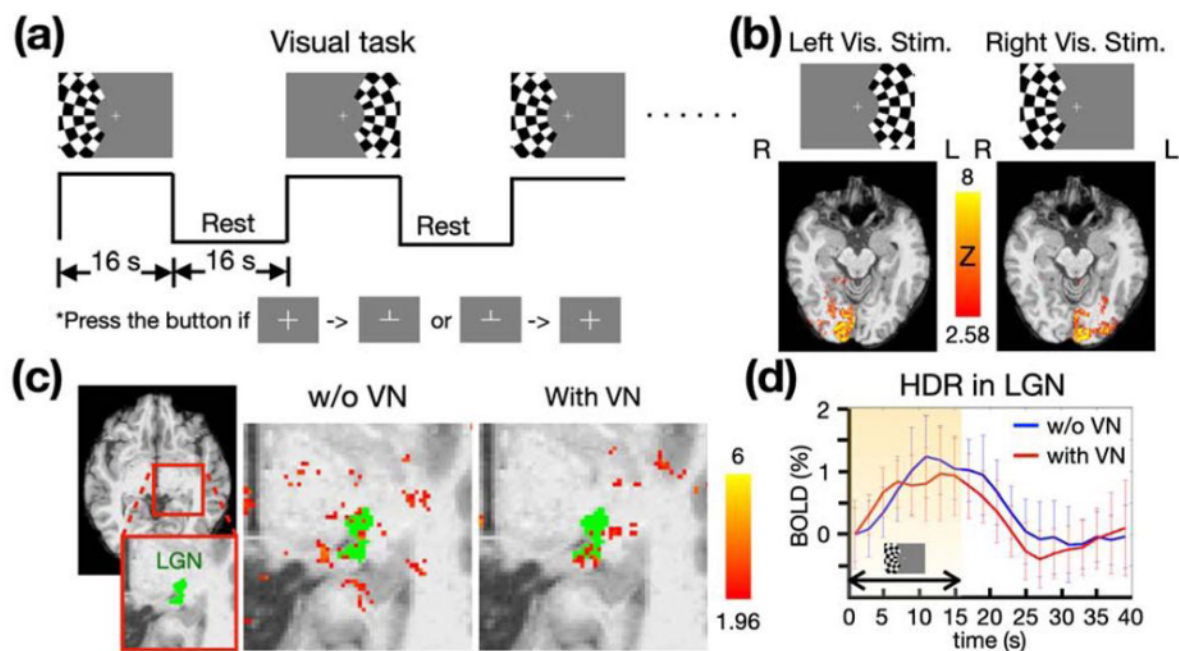


Figure 6

The visual task paradigm and the corresponding results from a single participant. (a) The checkerboard flashed alternately on the left and right sides for 16 seconds, followed by a 16-second resting period. (b) The BOLD activation maps correspond to the visual stimuli in the left or right visual hemifield. (c) The BOLD activation maps in the LGN, presented with and without a VN gradient. The z-statistics are overlaid on the individual T1 image and color-coded as indicated by the color bar. (d) The HDRs in the LGN, compared without and with a VN gradient (b value of the VN gradient is 30). The shaded area highlights the period of visual stimulation. The blue and red traces denote the mean HDRs of BOLD signal change obtained using the protocols without and with VN gradient, respectively. The whiskers denote the standard deviation across trials.

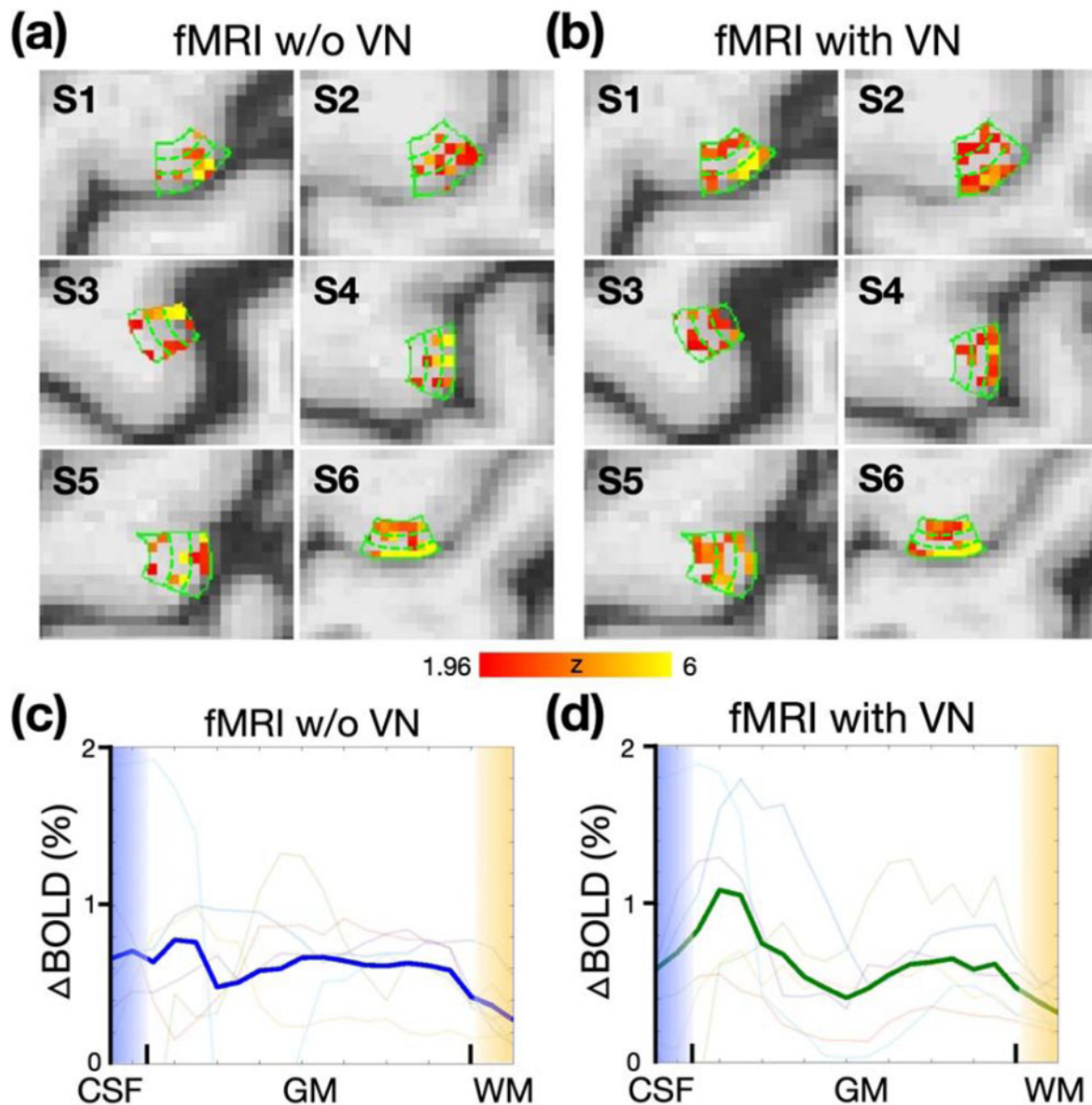


Figure 7

Activation maps and depth-dependent profiles of BOLD responses across participants. (a) BOLD activation maps in the absence of a VN gradient, with the solid green contour delineating the primary motor cortex (M1). Dashed traces delineate superficial, middle, and deep cortical layers. (b) Activation maps with a VN gradient applied. (c) Depth-dependent profiles of BOLD response in the absence of a VN. The dark blue traces denote the group-averaged data. (d) Depth-dependent profiles with VN applied. The dark red traces denote the group-averaged profiles with the VN gradient applied. In both (c) and (d), the traces with light colors indicate individual profiles.

Brain-wide directed functional connectivity

The group-averaged Fisher's z-transformed FC matrices are presented in **Figure 8a**, with ROIs organized according to their respective functional networks. The FC matrix on the left was derived from conventional fMRI data without the application of VN gradients, while the matrix on the right was generated using VN fMRI data. The group mean of Fisher's z values exhibited comparable outcomes between conventional fMRI and VN fMRI datasets. In **Figure 8b**, the group-level statistics for the directed functional connectivity matrix are presented, including the Visual (Vis), Sensorimotor (SM), Frontoparietal (FP), and Default-Mode Networks (DMN). The lower triangular part of the matrix displays uncorrected t-values, as derived from independent t-tests. The upper triangular part shows only the t-values that survived the False Discovery Rate (FDR) correction procedure (corrected $p < 0.001$) (Benjamini and Hochberg 1995; Benjamini and Yekutieli 2001). In the absence of a VN gradient, scarcely any of the FCs remained statistically significant following FDR correction, with minor exceptions within the visual network. Conversely, when VN fMRI is utilized, all examined functional networks exhibited statistically significant FCs. These findings suggest that VN fMRI enhances statistical power by effectively reducing inter-participant variability.

The layer specificity also enables the directed brain connectome as demonstrated in **Figure 8c**. Examples from each functional network are presented, specifically focusing on those ROI pairs that survived FDR correction in the VN fMRI dataset. Directed FCs of interest are denoted by red circles. Within the visual network, the primary visual cortex (V1) exhibited feedforward connectivity originating from its superficial layers and targeting the middle layers of the visual area V4. In the sensorimotor network, the M1 received inputs exclusively in its superficial layers, which originated from both the superficial and deep layers of the primary sensory cortex (S1). In the frontoparietal network, a reciprocal feed-forward relationship was observed between the left inferior frontal cortex (L-IFC) and the right inferior frontal cortex (R-IFC), originating predominantly from superficial to middle layers. In the default mode network (DMN), the posterior cingulate cortex (PCC) exhibited connectivity dominated by middle layers. The precuneus (PCu) also showed feedforward connectivity from PCC's superficial to its own middle layers, which is consistent with its role as a major hub in the DMN.

Discussion

To develop brain-wide directed functional connectivity using depth-dependent functional information, several technical criteria must be met: sub-mm spatial resolution, high spatial specificity, adequate temporal resolution, sufficient functional sensitivity, and global brain coverage. In this study, we addressed those challenges by incorporating a number of techniques, such as velocity-nulling gradients, SMS acceleration and NORDIC denoising techniques. As a result, we were able to develop a VN fMRI sequence featuring 0.9-mm isotropic spatial resolution, a TR of 3.84 seconds and brain-wide coverage. The effects of various parameters, including the TE and the b value associated with the VN gradient, were systematically evaluated. Our VN fMRI showcased its spatial specificity by revealing double-peak patterns within M1 during a finger-tapping task. Compared to conventional fMRI approaches, the VN fMRI exhibited greater robustness across participants. Furthermore, by properly selecting the b value of the VN gradient based on empirical evidence, our VN fMRI was capable of detecting BOLD signals not only at the cortical surface but also in subcortical regions, specifically the LGN. This optimized imaging framework enables the detailed examination of directed functional connectivity, demonstrating several layer-specific functional relationships consistent with the existing literature.

In 7T layer-dependent fMRI, bipolar diffusion gradients have been employed to mitigate the draining-vein effect, but in conjunction with T2 preparation methods (L. Huber, Hua, et al. 2017) rather rapid GE-EPI sequences. This is likely due to the limited TE associated with GE-EPI at 7T,

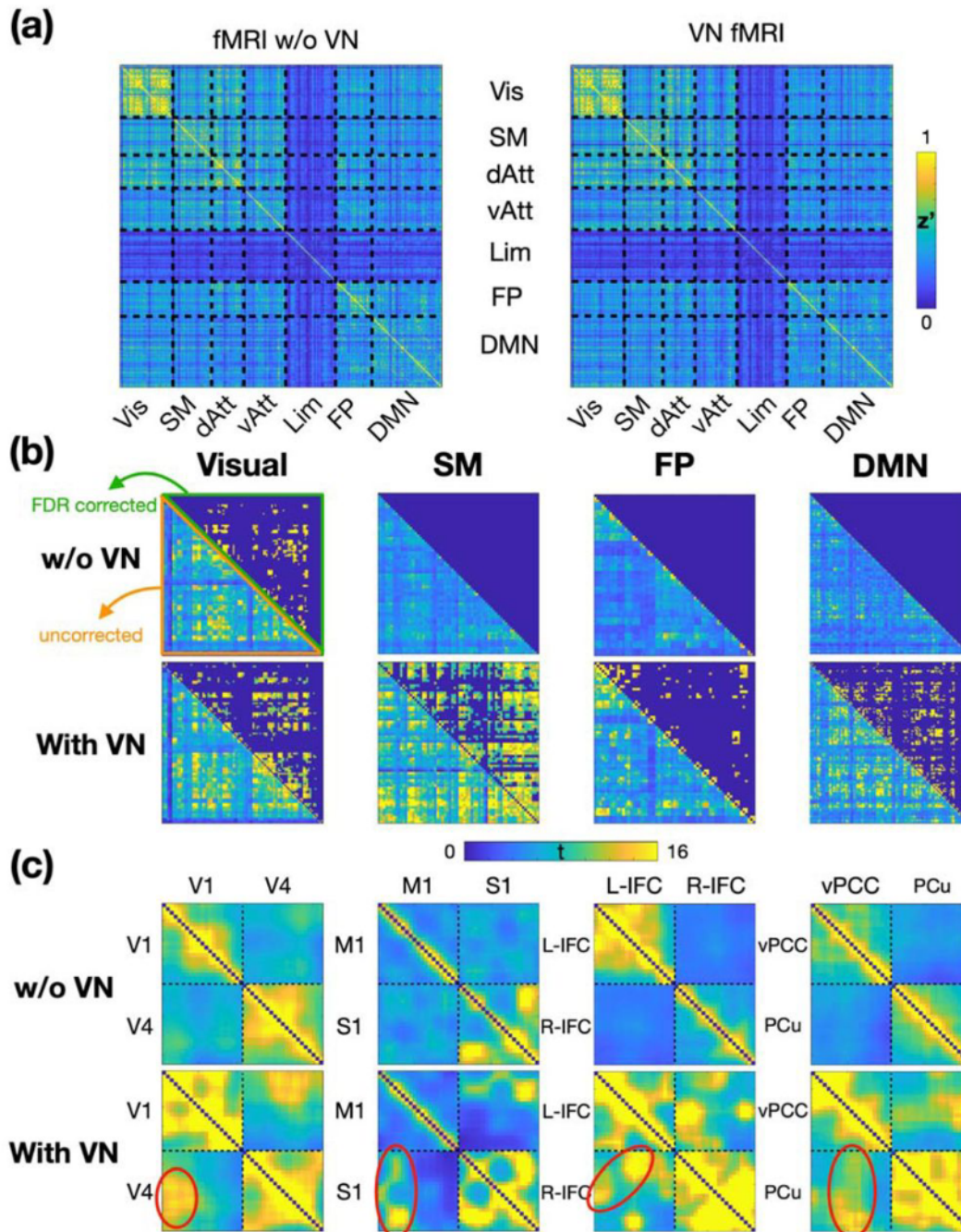


Figure 8

Directed functional connectivity across functional networks (N = 6). (a) Fisher's z transformed FC matrices. The FC matrix on the left represents the Fisher's z values without a VN gradient, while the matrix on the right corresponds to the Fisher's z values with a VN gradient. (b) Statistical results of FC within the visual, sensorimotor, frontoparietal, and default-mode networks. The matrix's lower triangular parts show uncorrected t-values, while the upper triangular parts show t-values that survived FDR correction (corrected $p < 0.001$). (c) Depth-dependent FC matrices for a representative ROI pair in each network. T-values are color-coded as indicated by the color bar. Red circles highlight directed functional connectivities of particular interest. The upper row displays the results without a VN gradient, and the lower row presents the results with a VN gradient. Abbreviations: z' – Fisher's z; M1 – primary motor cortex; S1 – primary sensory cortex; L-IFC – left inferior frontal cortex; R-IFC – right inferior frontal cortex; vPCC – ventral posterior cingulate cortex; PCu – precuneus; Vis – visual network; SM – sensorimotor network; dAtt – dorsal attention network; vAtt – ventral attention network; Lim – limbic network; FP – frontoparietal network; DMN – default-mode network.

which is insufficient to accommodate bipolar diffusion gradients in an ultra-high-resolution regime. However, it is noteworthy that the functional sensitivity of diffusion-weighted T2 preparation methods is considerably lower than the GE-BOLD method (L. Huber et al. 2021a [↗](#)). While researchers have explored higher magnetic field strengths to enhance sensitivity, the current study introduces a depth-dependent GE-EPI approach implemented at 3T. While this may initially appear counterintuitive, the application of GE-EPI at 3T affords several unique advantages. First, the extended T2* values at 3T lead to diminished T2* blurring along the phase-encoding direction, thus yielding a more focused point-spread function (PSF). Second, geometric distortions and signal losses due to susceptibility effects are less prominent at 3T relative to those at 7T. Third, the feasibility of conducting layer-dependent fMRI at 3T broadens the scope for multi-site, large-scale investigations. These attributes are of critical importance for a broad range of neuroscience applications.

To date, only a limited number of studies have reported on 3T layer-specific GE-BOLD fMRI (Koopmans, Barth, and Norris 2010 [↗](#); Scheeringa et al. 2016; Irati Markuerkiaga et al. 2021 [↗](#); Knudsen et al. 2023 [↗](#)). Among these, the study by Knudsen et al. (2023) [↗](#) addressed the draining-vein contamination issue by employing phase regression techniques to mitigate bias toward superficial cortical layers. Phase regression operates by removing signals from vessels that generate a non-zero net phase, typically vessels that are large and oriented in a particular direction. As illustrated in **Figure 5** [↗](#), the efficacy of phase regression diminishes slightly at a TE of 38 ms and becomes almost negligible at a TE of 43 ms. This reduction in efficacy suggests a minimal contribution from macrovascular sources to the BOLD signal at extended TEs. In contrast, **Figure 4d** [↗](#) demonstrates that VN gradients are still effective even at a long TE, likely attributable to the suppression of smaller, cortically-penetrating veins. Our findings suggest that while a phase regression method minimizes the macrovascular contribution to BOLD signal, VN gradients further suppress the signals from smaller, cortically-penetrating veins.

Although the VN gradients suppress the intravascular signal, the gradient itself does not address the extravascular effect that may influence the BOLD signals in middle and deep cortical layers, potentially causing vascular blurring across cortical layers. However, the extended TE due to the insertion of a VN gradient may alleviate this issue. Our results in **Figure 5** [↗](#) suggest that extended TE has an impact similar to that of the phase regression approach which has been shown to reduce both intravascular and extravascular contributions from pial veins (Stanley et al. 2021 [↗](#)). The extent to which the extravascular contribution is reduced warrants further investigation. Notably, in some double-peak activation patterns observed in this study, the BOLD amplitude corresponding to superficial layers substantially exceeded that related to deeper layers. This discrepancy, not evident in electrophysiological data (Maier 2010 [↗](#)), suggests potential residual extravascular effects. Therefore, caution is advised when interpreting BOLD amplitude as a direct indicator of neural activity strength. This consideration is less pertinent in layer-specific FC analyses due to the normalization of amplitude in the computation of correlation coefficients.

Beyond the use of GE-BOLD fMRI at a 3T magnetic field strength, the MAGEC VASO sequence has been shown to achieve both spatial specificity and volumetric brain coverage (~83 mm thickness), albeit with a relatively long TR of 8.2 seconds. Importantly, the energy spectrum of resting-state signals generally falls within 0.01 to 0.1 Hz range. According to the Nyquist sampling theorem, a TR shorter than 5 seconds is required to avoid temporal aliasing. Exceeding this limit can lead to frequency aliasing, wherein signals at the higher end of the bandwidth overlap with signals in other bands, thereby complicating the interpretation of resting-state FC. Our VN fMRI approach, featuring a TR of less than 4 seconds, effectively alleviates this issue of temporal aliasing.

Leveraging the ultrahigh-resolution, brain-wide coverage and layer specificity provided by the proposed VN fMRI, our results of directed FC in **Figure 8c** [↗](#) exhibit several established feedforward and feedback neural pathways. Within the visual network, our results demonstrate feedforward signal propagation from the superficial neural layers of the primary visual cortex

(V1) to the intermediate layers in the visual area 4 (V4). This feedforward connection is a component in the ventral stream of visual pathways (Goodale and Milner 1992; Kravitz et al. 2013). Additionally, within the sensorimotor network, we observed directed FC originating from the primary somatosensory cortex (S1) and targeting the superficial neural layers of M1. The FC of the posterior cingulate cortex (PCC) was predominantly observed in the middle laminae. These observations are in alignment with previously reported findings from a laminar-specific fMRI study (L. Huber, Handwerker, et al. 2017).

Despite the unprecedented features of our VN fMRI, this study is not without limitations, outlined as follows. 1) Residual intravascular signal: the VN gradients suppress signals originating from vascular flow oriented in specific directions. However, when vasomotion lacks coherence, the effective pseudo-diffusion coefficient (D^*) is reduced, thereby diminishing the efficiency of intravascular signal suppression. In **Figure 3**, **Figure 4** and **Figure 5**, residual intravascular signals are predominantly localized around regions of the pial surface exhibiting elevated curvature. This is likely attributable to incoherent flow at the branching junctures of pial vessels. Notably, these residual intravascular signals are generally confined to the vascular space, thus exerting minimal impact on the depth-dependent signal characteristics within the grey matter. 2) Inadequate TR for event-related paradigms: Although our VN fMRI acquisition reduced the TR from 8.2 s to 3.84 s, this TR duration remains suboptimal for studies employing event-related task designs. Future research would benefit from enhanced acceleration capabilities to facilitate the investigation of high-level cognitive processes. Such developments, however, are beyond the scope of this study.

Conclusion

In summary, the developed VN fMRI exhibited layer specificity and BOLD sensitivity successfully at 3T. Leveraging its brain-wide coverage and reasonable scan time, the VN fMRI yielded promising results in the study of directed FC. Given the widespread accessibility of 3T scanners, the potential impact of this development is expected to be extensive across various domains of neuroscience research.

Materials and Methods

Participants

In this study, we recruited a total of six healthy adults (2 females, age = 32.3 ± 8.5 years). All participants underwent screening to confirm no history of neurological or psychiatric conditions, previous head trauma, or MRI contraindications. Only right-handed individuals are included in this study. Prior to participation, each individual provided informed consent in accordance with the experimental protocol approved by the University of North Carolina at Chapel Hill Institutional Review Board.

Stimulation paradigms

All visual stimuli across the paradigms were presented using PsychoPy software (Version 2022.2.4) (Brainard 1997). The stimuli were displayed on a screen positioned at the head end of the magnet bore, and participants viewed the visual presentations through a mirror mounted on the head coil.

Finger-tapping task

In a finger tapping task, the primary motor cortex (M1) receives incoming sensory and associative information in its superficial layers. This activation subsequently propagates to the deeper layers, where output signals are generated to ultimately control finger muscles. Consequently, both the superficial and deep layers of M1 exhibit activation, while the middle layers typically do not. Such a double-peak response pattern at M1 has been used as a hallmark to evaluate the spatial specificity in layer-dependent fMRI studies (L. Huber, Handwerker, et al. 2017).

During the experiment, participants were instructed to perform a pinch-like tapping action between the thumb and index finger of their right hand. The timing and frequency of the finger tapping was synchronized with a video displayed on a screen within the scanner. The video consisted of a block-designed paradigm, as illustrated in [Figure 2a](#). The tapping frequency was 2 Hz with each 'ON' block spanning 30 seconds, immediately followed by a 30-second rest during the 'OFF' block. Depending on the repetition time (TR), the total number of blocks varied between 12 and 18. The overall numbers of fMRI volumes in finger-tapping tasks were maintained at either 282 or 286.

Visual task

The design of visual experiment is illustrated in [Figure 6a](#). The visual stimulus consisted of a high-contrast visual checkerboard, with its contrast reversing at a frequency of 8 Hz. The checkerboard alternately flashed on the left or right for 16 seconds, followed by a 16-second rest period. Participants were instructed to press a button on a response box when the crosshair changed from '+' to '⊥' or from '⊥' to '+'. These responses were recorded to verify the compliance to the instructions.

Common protocol in MR acquisitions

MR images were acquired using a Siemens 3T Prisma scanner (Siemens Healthcare, Erlangen, Germany) and a 32-channel head coil at the Biomedical Research Imaging Center (BRIC) at the University of North Carolina at Chapel Hill. For each participant, an MPRAGE image was acquired for structural imaging using the following imaging parameters: 0.8-mm isotropic resolution, TR/TE/TI = 2400/2.24/1060 ms, flip angle = 8°, in-plane acceleration factor = 2, partition thickness = 0.8 mm, 208 partitions, sagittal slicing, image matrix = 320 × 300, and FOV = 25.6 cm × 24.0 cm.

For functional imaging, the customized SMS sequence with VN gradient was implemented in the vendor-provided IDEA environment (VE11E). The functional data were acquired using a blipped-controlled aliasing in parallel imaging (blipped-CAIPI) SMS imaging (Setsompop et al. 2012) with two-dimensional (2D) single-shot EPI readout. While imaging protocols varied slightly across different experiments, they all shared the following parameters: isotropic spatial resolution of 0.9 mm; axial slicing; in-plane image dimensions of 224×210; frequency and phase encoding in the left-right and anterior-posterior directions, respectively; and an in-plane acceleration rate of 2. A 50-mm-thick saturation band was utilized to suppress the signal from the eyeballs. Additionally, a brief fMRI acquisition with an opposite phase-encoding direction was acquired for distortion correction.

Experiment configuration and imaging protocols in optimization phase

The data acquisitions in this study consisted of two phases: optimization and validation. During the optimization phase, various scan protocols were evaluated to establish the optimal scan parameters. To minimize inter-subject variation, data were solely collected from a single

participant. This phase included two experiments: 1) b-value optimization and 2) acceleration rate optimization.

Experiment 1A: Optimization of b value in VN gradient

Empirical tests were conducted with b values of 0, 15, 30 and 48, based on the simulation results shown in [Figure 1b](#). A finger-tapping task was employed to assess BOLD sensitivity and spatial specificity. To avoid confounds from reconstruction errors associated with a high acceleration rate at this preliminary stage, we acquired a slab encompassing the primary motor cortex (M1) without SMS acceleration, as shown in [Figure 2b](#). The slab placement was under the supervision of an on-site radiologist. With the acquisition of 21 slices in total, the resultant slab thickness was 18.9 mm. The TEs corresponding to these b values were 33, 41, 43, and 45 ms, respectively, while the TR was consistently maintained at 2.54 s. The total number of acquired volumes was 286. The total number of finger-tapping blocks was 12.

Experiment 1B: Optimization of SMS acceleration rate

Higher SMS acceleration rates can decrease the volumetric scan time of fMRI, which potentially enhances its statistical power and captures a broader range of temporal dynamics. However, a high acceleration rate might also compromise the quality of image reconstruction ([Cauley et al. 2014](#); [Todd et al. 2016](#)). The relationship between SMS acceleration and laminar activation profiles is still unclear. In contrast, the volumetric scan time of fMRI should not be excessively prolonged. Given that the majority of energy in the BOLD signal spectrum falls between 0.01 and 0.1Hz, it is preferable to ensure that the TR does not exceed 4 seconds. Hence, we tested SMS acceleration rates of 4, 5, and 6 with an in-plane acceleration rate of 2, resulting in TRs of 3.84, 3.11, and 2.63 seconds, respectively. The finger-tapping task was employed to assess the impacts on laminar activation profiles.

We conducted four tests using different scanning parameters: 1) a SMS factor of 4 without VN gradient, 2) a SMS factor of 4 with VN gradient, 3) a SMS factor of 5 with VN gradient, and 4) a SMS factor of 6 with VN gradient. The b value of VN gradient was set as 30 based on preliminary results from Experiment 1A. The TE and total number of acquired volumes were kept constant at 43 ms and 282, respectively. The numbers of slices per test were 124, 124, 125, and 126, respectively, and the total number of finger-tapping blocks per test were 12, 12, 14, and 18, respectively. The spatial shifts caused by CAIPI blips for SMS factor of 4, 5, 6 were FOV/2, FOV/3 and FOV/3 respectively.

Experiment configuration and imaging protocols in validation phase

In the validation phase, the b value of VN gradient and SMS acceleration rate were predefined based on the results in the optimization phase. We examined 1) the impact of TE on depth-dependent profiles, 2) the BOLD sensitivity in subcortical regions, 3) the robustness of laminar activation profiles, and 4) the feasibility of brain-wide directed functional connectivity (FC).

Experiment 2A: Impact of TE on depth-dependent profiles

In conventional 3T fMRI, the typical TE ranges between 30-35 ms ([Van Essen et al. 2013](#)). However, to incorporate the VN gradient, the TE in VN fMRI is extended. While the VN gradient mitigates the draining-vein effect, it is noteworthy that the intravascular and extravascular contributions to the BOLD signal also vary with changes in TE ([Duong et al. 2003](#); [Kim and Ogawa 2012](#); [Irati Markuerkiaga et al. 2021](#)). To evaluate the impact of this increased TE, a participant underwent multiple finger-tapping task sessions with TEs of 33, 38, and 43 ms, but without the VN gradient. For comparative analysis, the same participant also performed the finger-tapping task with both a TE of 43 ms and the VN gradient. In total, four fMRI sessions were conducted for a single

participant: three sessions were conducted without a VN gradient, and one session utilized a VN gradient with a predefined b value of 30 s/mm². All the sessions utilized an SMS factor of 4 and a repetition time (TR) of 3.84 seconds. Each session recorded a total of 282 volumes.

Experiment 2B: BOLD sensitivity in subcortical regions

Subcortical regions typically exhibit weaker blood-oxygen-level-dependent (BOLD) responses in comparison to cerebral cortical areas such as the primary motor or somatosensory cortices. This can be attributed to a variety of factors, including a lower SNR and decreased functional specificity. To ensure that the VN gradient does not result in substantial attenuation of subcortical BOLD signals, we investigated the BOLD response within the lateral geniculate nucleus (LGN) using a visual checkerboard task, as outlined in the section of Stimulation Paradigms. Two fMRI sessions were conducted for a single participant, one in the absence of a VN gradient and the other utilizing a VN gradient with a b value of 30. In each session, the participant was presented with twelve stimuli per hemifield, resulting in a total of 24 stimulation blocks. Both sessions utilized an SMS factor of 4 and a repetition time (TR) of 3.84 seconds. A total of 202 volumes were recorded for each session.

Experiment 2C: Robustness of laminar activation profile

Six participants were enrolled in this experiment. Each participant underwent two fMRI sessions of finger-tapping task. One in the absence of a VN gradient and the other utilizing a VN gradient with a b value of 30. The sequence of the two sessions was randomized across participants. For each session, a total of 282 volumes were acquired. The SMS factor and TR were the same as those used in Experiment 2A.

Experiment 2D: Feasibility of brain-wide directed functional connectivity

Resting-state fMRI data were collected from six participants. Each participant underwent two resting-state fMRI sessions. Participants were instructed to stay motionless, keep their eyes closed, and avoid falling asleep. Only one of the sessions employed the VN gradient with a b value of 30, with the sequence of the two sessions being randomized among participants. The SMS factor and TR were the same as those used in Experiment 2A. For each session, a total of 300 volumes were acquired, taking approximately 19.5 minutes.

Data Preprocessing

The SMS-accelerated EPI time series were reconstructed using in-house MATLAB code, which performed slice Generalized Autocalibrating Partial Parallel Acquisition (slice-GRAPPA) (Setsompop et al. 2012) and in-plane GRAPPA (Griswold et al. 2002) jointly in one step. The reconstructed images were denoised using NORDIC method with estimated g -factor map (function name: 'NIFTI_NORDIC'). The input images were complex-valued. The width of the smoothing filter for the phase was set as 10, while all other hyperparameters retained their default values. After NORDIC denoising, the images were partial-Fourier reconstructed using a projection-onto-convex-set (POCS) method (Haacke, Lindskog, and Lin 1991). The time-series data were then motion corrected using FSL, slice-timing corrected using sinc interpolation, and distortion corrected using FSL TOPUP (Smith et al. 2004). For artifact removal, we decomposed the magnitude image of the corrected data into a number of independent components by MELODIC (Beckmann et al. 2005). The noise components were labeled manually and then removed using the FSL command 'fsl_regfilt'. The time-series signals were band-pass filtered from 0.01 to 0.125 Hz.

Phase regression

This study implemented the phase regression method for comparative purposes. The phase regression approach was applied to remove the macrovascular component in the BOLD signal as indicated in previous reports (Menon 2002; Knudsen et al. 2023). This macrovascular

component not only contributes to alterations in signal magnitude, but also introduces phase variations. By utilizing a linear regression approach between the time series of signal magnitude and phase, the macrovascular-associated contribution to the BOLD signal could be suppressed.

The phase time series underwent the same data processing steps as outlined above except for the step of artifact removal. The phase images of the distortion corrected data were linearly regressed against the time courses of the identified noise components. We employed linear regression for phase images because the FSL command ‘fsl_regfilt’ is incompatible with complex-valued images. Lastly, the filtered BOLD time series of signal magnitude were linearly regressed using the fully-processed and phase-unwrapped time series as the covariate.

Depth-dependent statistical analysis in volume space

The event-related responses were analyzed using FSL FEAT (M. W. Woolrich et al. 2001 [↗](#); Mark W. Woolrich, Behrens, and Smith 2004 [↗](#)). FLAME 1 was used for group-level analysis (Mark W. Woolrich et al. 2004 [↗](#)). For the depth-dependent analysis of the finger-tapping task, the M1 area of each individual was identified based on the BOLD activation map. The subsampling along cortical depth can be either volume-based or surface-based. For volume-based subsampling, we generally follow the pipeline suggested by Dr. Huber (<https://layerfmri.com/2018/03/11/quick-layering/#more-531> [↗](#)). We up-sampled the image slice containing M1 by 5 times, thus achieving an in-plane resolution of 0.18 mm isotropic. The interpolation method utilized was the nearest neighbor. The boundaries between grey matter/white matter (GM/WM) and cerebral spinal fluid/grey matter (CSF/GM) within M1 were manually delineated. With the created M1 label, we employed the LayNii software tool to form 20 equidistant layers (L. R. Huber et al. 2021 [↗](#)). The number 20 was determined according to the suggestion by LayNii that the number of layers may be set to at least 4 times larger than the resolution. The depth-dependent functional data were extracted based on the calculated cortical depths. Finally, the depth-dependent profile was computed by averaging the depth-dependent functional data within each layer.

Surface-based directed functional connectivity analysis

For surface-based subsampling along the cortical depth, we linearly coregistered the individual T1 volume onto the time-averaged EPI volume using Advanced Normalization Tools (ANTs; <http://stnava.github.io/ANTs/> [↗](#)). Following this, we resampled the processed EPI volumes onto the cortical surface at different cortical depths using the FreeSurfer command ‘mri_vol2surf’. Adapting the layering strategy in volume space, we converted 20 cortical layers from volume to surface space. Here, the cortical depth ranged from -0.125 to 1.0625, with 0 and 1 representing the GM/WM and CSF/GM boundaries, respectively. Surface-based spatial smoothing was applied with full-width half-magnitude (FWHM) of 3 mm.

In order to investigate directed functional connectivity throughout the brain, we employed the Shen268 functional parcellation (Shen et al. 2013 [↗](#)). The volume-based Shen268 parcellation was converted onto the cortical surface as shown in **Figure 9a** [↗](#). The volumetric EPI time-series for each individual were resampled onto their individual brain surfaces before being projected onto the template brain surface. The template surface is of ‘fsaverage5’ from FreeSurfer. We also categorized the parcels in Shen268 atlas into 7 functional networks using the functional atlas reported by Yeo et al. (Thomas Yeo et al. 2011 [↗](#)). The time course associated with each region-of-interest (ROI) was obtained by averaging all time courses within the ROI. The depth-dependent functional connectivity was calculated using Pearson correlation and then converted into Fisher’s z values, as illustrated in **Figure 9b** [↗](#).

For the purposes of group-level statistical analysis, depth-dependent layers were initially partitioned into superficial, middle, and deep layers, as depicted in the left panel of **Figure 9c** [↗](#). Within each block of paired layers, Fisher’s z-values were averaged, thereby reducing the 20-by-20 functional connectivity (FC) matrix for each ROI pair to a simplified 3-by-3 matrix. This size

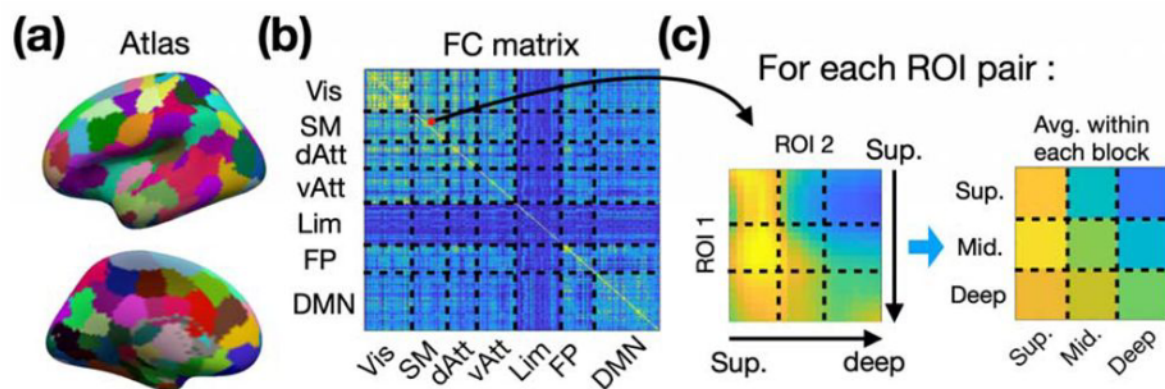


Figure 9

Schematic Overview of Directed Functional Connectivity Calculation Procedure. (a) The surface-based Shen268 functional parcellation. (b) The depth-dependent functional connectivity matrix. (c) The depth-dependent layers were separated into superficial, middle, and deep. The functional connectivity (FC) matrix for each ROI pair was reduced to a simplified 3-by-3 matrix for multiple comparison. Abbreviations: Vis – visual network; SM – sensorimotor network; dAtt – dorsal attention network; vAtt – ventral attention network; Lim – limbic network; FP – frontoparietal network; DMN – default-mode network.

reduction facilitated the management of multiple comparisons. Multiple comparison corrections were performed using the Benjamini & Hochberg False Discovery Rate (FDR) procedure (Benjamini and Hochberg 1995 [DOI](#)).

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Article and author information

Wei-Tang Chang

Biomedical Research Imaging Center, University of North Carolina at Chapel Hill, NC, USA, Department of Radiology, University of North Carolina at Chapel Hill, NC, USA, Department of Biomedical Engineering, University of North Carolina at Chapel Hill, NC, USA

For correspondence: weitang_chang@med.unc.edu

Weili Lin

Biomedical Research Imaging Center, University of North Carolina at Chapel Hill, NC, USA, Department of Radiology, University of North Carolina at Chapel Hill, NC, USA

Kelly S. Giovanello

Biomedical Research Imaging Center, University of North Carolina at Chapel Hill, NC, USA, Department of Psychology & Neuroscience, University of North Carolina at Chapel Hill, NC, USA

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Reviewer #1 (Public Review):

Summary:

This study aims to provide imaging methods for users of the field of human layer-fMRI. This is an emerging field with 240 papers published so far. Different than implied in the manuscript, 3T is well represented among those papers. E.g. see the papers below that are not cited in the manuscript. Thus, the claim on the impact of developing 3T methodology for wider dissemination is not justified. Specifically, because some of the previous papers perform whole brain layer-fMRI (also at 3T) in more efficient, and more established procedures.

The authors implemented a sequence with lots of nice features. Including their own SMS EPI, diffusion bipolar pulses, eye-saturation bands, and they built their own reconstruction around it. This is not trivial. Only a few labs around the world have this level of engineering expertise. I applaud this technical achievement. However, I doubt that any of this is the right tool for layer-fMRI, nor does it represent an advancement for the field. In the thermal noise dominated regime of sub-millimeter fMRI (especially at 3T), it is established to use 3D readouts over 2D (SMS) readouts. While it is not trivial to implement SMS, the vendor implementations (as well as the CMRR and MGH implementations) are most widely applied across the majority of current fMRI studies already. The author's work on this does not serve any previous shortcomings in the field.

The mechanism to use bi-polar gradients to increase the localization specificity is doubtful to me. In my understanding, killing the intra-vascular BOLD should make it less specific. Also, the empirical data do not suggest a higher localization specificity to me.

Embedding this work in the literature of previous methods is incomplete. Recent trends of vessel signal manipulation with ABC or VAPER are not mentioned. Comparisons with VASO are outdated and incorrect.

The reproducibility of the methods and the result is doubtful (see below).

I don't think that this manuscript is in the top 50% of the 240 layer-fmri papers out there.

3T layer-fMRI papers that are not cited:

Taso, M., Munsch, F., Zhao, L., Alsop, D.C., 2021. Regional and depth-dependence of cortical blood-flow assessed with high-resolution Arterial Spin Labeling (ASL). *Journal of Cerebral Blood Flow and Metabolism*. <https://doi.org/10.1177/0271678X20982382>

Wu, P.Y., Chu, Y.H., Lin, J.F.L., Kuo, W.J., Lin, F.H., 2018. Feature-dependent intrinsic functional connectivity across cortical depths in the human auditory cortex. *Scientific Reports* 8, 1-14. <https://doi.org/10.1038/s41598-018-31292-x>

- Lifshits, S., Tomer, O., Shamir, I., Barazany, D., Tsarfaty, G., Rosset, S., Assaf, Y., 2018. Resolution considerations in imaging of the cortical layers. *NeuroImage* 164, 112-120. <https://doi.org/10.1016/j.neuroimage.2017.02.086>
- Puckett, A.M., Aquino, K.M., Robinson, P.A., Breakspear, M., Schira, M.M., 2016. The spatiotemporal hemodynamic response function for depth-dependent functional imaging of human cortex. *NeuroImage* 139, 240-248. <https://doi.org/10.1016/j.neuroimage.2016.06.019>
- Olman, C.A., Inati, S., Heeger, D.J., 2007. The effect of large veins on spatial localization with GE BOLD at 3 T: Displacement, not blurring. *NeuroImage* 34, 1126-1135. <https://doi.org/10.1016/j.neuroimage.2006.08.045>
- Ress, D., Glover, G.H., Liu, J., Wandell, B., 2007. Laminar profiles of functional activity in the human brain. *NeuroImage* 34, 74-84. <https://doi.org/10.1016/j.neuroimage.2006.08.020>
- Huber, L., Kronbichler, L., Stirnberg, R., Ehse, P., Stocker, T., Fernández-Cabello, S., Poser, B.A., Kronbichler, M., 2023. Evaluating the capabilities and challenges of layer-fMRI VASO at 3T. *Aperture Neuro* 3. <https://doi.org/10.52294/001c.85117>
- Scheeringa, R., Bonnefond, M., van Mourik, T., Jensen, O., Norris, D.G., Koopmans, P.J., 2022. Relating neural oscillations to laminar fMRI connectivity in visual cortex. *Cerebral Cortex*. <https://doi.org/10.1093/cercor/bhac154>

Strengths:

See above. The authors developed their own SMS sequence with many features. This is important to the field. And does not leave sequence development work to view isolated monopoly labs. This work democratizes SMS. The questions addressed here are of high relevance to the field: getting tools with good sensitivity, user-friendly applicability, and locally specific brain activity mapping is an important topic in the field of layer-fMRI.

Weaknesses:

1. I feel the authors need to justify why flow-crushing helps localization specificity. There is an entire family of recent papers that aim to achieve higher localization specificity by doing the exact opposite. Namely, MT or ABC fMRI aims to increase the localization specificity by highlighting the intravascular BOLD by means of suppressing non-flowing tissue. To name a few:

Priovoulos, N., de Oliveira, I.A.F., Poser, B.A., Norris, D.G., van der Zwaag, W., 2023. Combining arterial blood contrast with BOLD increases fMRI intracortical contrast. *Human Brain Mapping* hbm.26227. <https://doi.org/10.1002/hbm.26227>.

Pfaffenrot, V., Koopmans, P.J., 2022. Magnetization Transfer weighted laminar fMRI with multi-echo FLASH. *NeuroImage* 119725. <https://doi.org/10.1016/j.neuroimage.2022.119725>

Schulz, J., Fazal, Z., Metere, R., Marques, J.P., Norris, D.G., 2020. Arterial blood contrast (ABC) enabled by magnetization transfer (MT): a novel MRI technique for enhancing the measurement of brain activation changes. *bioRxiv*. <https://doi.org/10.1101/2020.05.20.106666>

Based on this literature, it seems that the proposed method will make the vein problem worse, not better. The authors could make it clearer how they reason that making GE-BOLD signals more extra-vascular weighted should help to reduce large vein effects.

The empirical evidence for the claim that flow crushing helps with the localization specificity should be made clearer. The response magnitude with and without flow crushing looks pretty much identical to me (see Fig, 6d).

It's unclear to me what to look for in Fig. 5. I cannot discern any layer patterns in these maps. It's too noisy. The two maps of TE=43ms look like identical copies from each other. Maybe an editorial error?

The authors discuss bipolar crushing with respect to SE-BOLD where it has been previously applied. For SE-BOLD at UHF, a substantial portion of the vein signal comes from the intravascular compartment. So I agree that for SE-BOLD, it makes sense to crush the intravascular signal. For GE-BOLD however, this reasoning does not hold. For GE-BOLD (even at 3T), most of the vein signal comes from extravascular dephasing around large unspecific veins, and the bipolar crushing is not expected to help with this.

2. The bipolar crushing is limited to one single direction of flow. This introduces a lot of artificial variance across the cortical folding pattern. This is not mentioned in the manuscript. There is an entire family of papers that perform layer-fMRI with black-blood imaging that solves this with a 3D contrast preparation (VAPER) that is applied across a longer time period, thus killing the blood signal while it flows across all directions of the vascular tree. Here, the signal crushing is happening with a 2D readout as a "snap-shot" crushing. This does not allow the blood to flow in multiple directions.

VAPER also accounts for BOLD contaminations of larger draining veins by means of a tag-control sampling. The proposed approach here does not account for this contamination.

Chai, Y., Li, L., Huber, L., Poser, B.A., Bandettini, P.A., 2020. Integrated VASO and perfusion contrast: A new tool for laminar functional MRI. *NeuroImage* 207, 116358. <https://doi.org/10.1016/j.neuroimage.2019.116358>

Chai, Y., Liu, T.T., Marrett, S., Li, L., Khojandi, A., Handwerker, D.A., Alink, A., Muckli, L., Bandettini, P.A., 2021. Topographical and laminar distribution of audiovisual processing within human planum temporale. *Progress in Neurobiology* 102121. <https://doi.org/10.1016/j.pneurobio.2021.102121>

If I would recommend anyone to perform layer-fMRI with blood crushing, it seems that VAPER is the superior approach. The authors could make it clearer why users might want to use the unidirectional crushing instead.

3. The comparison with VASO is misleading.

The authors claim that previous VASO approaches were limited by TRs of 8.2s. The authors might be advised to check the latest literature of the last years.

Koiso et al. performed whole brain layer-fMRI VASO at 0.8mm at 3.9 seconds (with reliable activation), 2.7 seconds (with unconvincing activation pattern, though), and 2.3 (without activation).

Also, whole brain layer-fMRI BOLD at 0.5mm and 0.7mm has been previously performed by the Juelich group at TRs of 3.5s (their TR definition is 'fishy' though).

Koiso, K., Müller, A.K., Akamatsu, K., Dresbach, S., Gulban, O.F., Goebel, R., Miyawaki, Y., Poser, B.A., Huber, L., 2023. Acquisition and processing methods of whole-brain layer-fMRI VASO and BOLD: The Kenshu dataset. *Aperture Neuro* 34. <https://doi.org/10.1101/2022.08.19.504502>

Yun, S.D., Pais-Roldán, P., Palomero-Gallagher, N., Shah, N.J., 2022. Mapping of whole-cerebrum resting-state networks using ultra-high resolution acquisition protocols. *Human Brain Mapping*. <https://doi.org/10.1002/hbm.25855>

Pais-Roldan, P., Yun, S.D., Palomero-Gallagher, N., Shah, N.J., 2023. Cortical depth-dependent human fMRI of resting-state networks using EPIK. *Front. Neurosci.* 17, 1151544. <https://doi.org/10.3389/fnins.2023.1151544>

The authors are correct that VASO is not advised as a turn-key method for lower brain areas, incl. Hippocampus and subcortex. However, the authors use this word of caution that is intended for inexperienced "users" as a statement that this cannot be performed. This statement is taken out of context. This statement is not from the academic literature. It's advice for the 40+ user base that wants to perform layer-fMRI as a plug-and-play routine tool in neuroscience usage. In fact, sub-millimeter VASO is routinely being performed by MRI-physicists across all brain areas (including deep brain structures, hippocampus etc). E.g. see Koiso et al. and an overview lecture from a layer-fMRI workshop that I had recently attended: <https://youtu.be/kzh-nWXd54s?si=hoIjLLIxFUJ4g20&t=2401>

Thus, the authors could embed this phrasing into the context of their own method that they are proposing in the manuscript. E.g. the authors could state whether they think that their sequence has the potential to be disseminated across sites, considering that it requires slow offline reconstruction in Matlab?

Do the authors think that the results shown in Fig. 6c are suggesting turn-key acquisition of a routine mapping tool? In my humble opinion, it looks like random noise, with most of the activation outside the ROI (in white matter).

4. The repeatability of the results is questionable.

The authors perform experiments about the robustness of the method (line 620). The corresponding results are not suggesting any robustness to me. In fact, the layer profiles in Fig. 4c vs. Fig 4d are completely opposite. The location of peaks turns into locations of dips and vice versa.

The methods are not described in enough detail to reproduce these results.

The authors mention that their image reconstruction is done "using in-house MATLAB code" (line 634). They do not post a link to github, nor do they say if they share this code.

It is not trivial to get good phase data for fMRI. The authors do not mention how they perform the respective coil-combination.

No data are shared for reproduction of the analysis.

5. The application of NODRIC is not validated.

Previous applications of NORDIC at 3T layer-fMRI have resulted in mixed success. When not adjusted for the right SNR regime it can result in artifactual reductions of beta scores, depending on the SNR across layers. The authors could validate their application of NORDIC and confirm that the average layer-profiles are unaffected by the application of NORDIC. Also, the NORDIC version should be explicitly mentioned in the manuscript.

Akbari, A., Gati, J.S., Zeman, P., Liem, B., Menon, R.S., 2023. Layer Dependence of Monocular and Binocular Responses in Human Ocular Dominance Columns at 7T using VASO and BOLD (preprint). Neuroscience. <https://doi.org/10.1101/2023.04.06.535924>

Knudsen, L., Guo, F., Huang, J., Blicher, J.U., Lund, T.E., Zhou, Y., Zhang, P., Yang, Y., 2023. The laminar pattern of proprioceptive activation in human primary motor cortex. bioRxiv. <https://doi.org/10.1101/2023.10.29.564658>

- <https://doi.org/10.7554/eLife.92805.1.sa2>

Reviewer #2 (Public Review):

This study developed a setup for laminar fMRI at 3T that aimed to get the best from all worlds in terms of brain coverage, temporal resolution, sensitivity to detect functional responses, and spatial specificity. They used a gradient-echo EPI readout to facilitate sensitivity, brain coverage and temporal resolution. The former was additionally boosted by NORDIC denoising and the latter two were further supported by parallel-imaging acceleration both in-plane and

across slices. The authors evaluated whether the implementation of velocity-nulling (VN) gradients could mitigate macrovascular bias, known to hamper the laminar specificity of gradient-echo BOLD.

The setup allows for 0.9 mm isotropic acquisitions with large coverage at a reasonable TR (at least for block designs) and the fMRI results presented here were acquired within practical scan-times of 12-18 minutes. Also, in terms of the availability of the method, it is favorable that it benefits from lower field strength (additional time for VN-gradient implementation, afforded by longer gray matter T2*).

The well-known double peak feature in M1 during finger tapping was used as a test-bed to evaluate the spatial specificity. They were indeed able to demonstrate two distinct peaks in group-level laminar profiles extracted from M1 during finger tapping, which was largely free from superficial bias. This is rather intriguing as, even at 7T, clear peaks are usually only seen with spatially specific non-BOLD sequences. This is in line with their simple simulations, which nicely illustrated that, in theory, intravascular macrovascular signals should be suppressible with only minimal suppression of microvasculature when small b-values of the VN gradients are employed. However, the authors do not state how ROIs were defined making the validity of this finding unclear; were they defined from independent criteria or were they selected based on the region mostly expressing the double peak, which would clearly be circular? In any case, results are based on a very small sub-region of M1 in a single slice - it would be useful to see the generalizability of superficial-bias-free BOLD responses across a larger portion of M1.

As repeatedly mentioned by the authors, a laminar fMRI setup must demonstrate adequate functional sensitivity to detect (in this case) BOLD responses. The sensitivity evaluation is unfortunately quite weak. It is mainly based on the argument that significant activation was found in a challenging sub-cortical region (LGN). However, it was a single participant, the activation map was not very convincing, and the demonstration of significant activation after considerable voxel-averaging is inadequate evidence to claim sufficient BOLD sensitivity. How well sensitivity is retained in the presence of VN gradients, high acceleration factors, etc., is therefore unclear. The ability of the setup to obtain meaningful functional connectivity results is reassuring, yet, more elaborate comparison with e.g., the conventional BOLD setup (no VN gradients) is warranted, for example by comparison of tSNR, quantification and comparison of CNR, illustration of unmasked-full-slice activation maps to compare noise-levels, comparison of the across-trial variance in each subject, etc. Furthermore, as NORDIC appears to be a cornerstone to enable submillimeter resolution in this setup at 3T, it is critical to evaluate its impact on the data through comparison with non-denoised data, which is currently lacking.

The proposed setup might potentially be valuable to the field, which is continuously searching for techniques to achieve laminar specificity in gradient echo EPI acquisitions. Nonetheless, the above considerations need to be tackled to make a convincing case.

- <https://doi.org/10.7554/eLife.92805.1.sa1>

Reviewer #3 (Public Review):

Summary:

The authors are looking for a spatially specific functional brain response to visualise non-invasively with 3T (clinical field strength) MRI. They propose a velocity-nulled weighting to remove the signal from draining veins in a submillimeter multiband acquisition.

Strengths:

- This manuscript addresses a real need in the cognitive neuroscience community interested in imaging responses in cortical layers in-vivo in humans.

- An additional benefit is the proposed implementation at 3T, a widely available field strength.

Weaknesses:

- Although the VASO acquisition is discussed in the introduction section, the VN-sequence seems closer to diffusion-weighted functional MRI. The authors should make it more clear to the reader what the differences are, and how results are expected to differ. Generally, it is not so clear why the introduction is so focused on the VASO acquisition (which, curiously, lacks a reference to Lu et al 2013). There are many more alternatives to BOLD-weighted imaging for fMRI. CBF-weighted ASL and GRASE have been around for a while, ABC and double-SE have been proposed more recently.
- The comparison in Figure 2 for different b-values shows % signal changes. However, as the baseline signal changes dramatically with added diffusion weighting, this is rather uninformative. A plot of t-values against cortical depth would be much more insightful.
- Surprisingly, the %-signal change for a b-value of 0 is not significantly different from 0 in the gray matter. This raises some doubts about the task or ROI definition. A finger-tapping task should reliably engage the primary motor cortex, even at 3T, and even in a single participant.
- The BOLD weighted images in Figure 3 show a very clear double-peak pattern. This contradicts the results in Figure 2 and is unexpected given the existing literature on BOLD responses as a function of cortical depth.
- Given that data from Figures 2, 3, and 4 are derived from a single participant each, order and attention affects might have dramatically affected the observed patterns. Especially for Figure 4, neither BOLD nor VN profiles are really different from 0, and without statistical values or inter-subject averaging, these cannot be used to draw conclusions from.
- In Figure 5, a phase regression is added to the data presented in Figure 4. However, for a phase regression to work, there has to be a (macrovascular) response to start with. As none of the responses in Figure 4 are significant for the single participant dataset, phase regression should probably not have been undertaken. In this case, the functional 'responses' appear to increase with phase regression, which is contra-intuitive and deserves an explanation.
- Consistency of responses is indeed expected to increase by a removal of the more variable vascular component. However, the microvascular component is always expected to be smaller than the combination of microvascular+macrovascular responses. Note that the use of %signal changes may obscure this effect somewhat because of the modified baseline. Another expected feature of BOLD profiles containing both micro- and macrovasculature is the draining towards the cortical surface. In the profiles shown in Figure 7, this is completely absent. In the group data, no significant responses to the task are shown anywhere in the cortical ribbon.
- Although I'd like to applaud the authors for their ambition with the connectivity analysis, I feel that acquisitions that are so SNR starved as to fail to show a significant response to a motor task should not be used for brain wide directed connectivity analysis.

The claim of specificity is supported by the observation of the double-peak pattern in the motor cortex, previously shown in multiple non-BOLD studies. However, this same pattern is shown in some of the BOLD weighted data, which seems to suggest that the double-peak pattern is not solely due to the added velocity nulling gradients. In addition, the well-known draining towards the cortical surface is not replicated for the BOLD-weighted data in Figures 3, 4, or 7. This puts some doubt about the data actually having the SNR to draw conclusions about the observed patterns.

- <https://doi.org/10.7554/eLife.92805.1.sa0>

Author Response:

Reviewer #1 (Public Review):

[...] Weaknesses:

1. I feel the authors need to justify why flow-crushing helps localization specificity. There is an entire family of recent papers that aim to achieve higher localization specificity by doing the exact opposite. Namely, MT or ABC fMRI aims to increase the localization specificity by highlighting the intravascular BOLD by means of suppressing non-flowing tissue. To name a few:

Priovoulos, N., de Oliveira, I.A.F., Poser, B.A., Norris, D.G., van der Zwaag, W., 2023. Combining arterial blood contrast with BOLD increases fMRI intracortical contrast. *Human Brain Mapping* hbm.26227. <https://doi.org/10.1002/hbm.26227>.

Pfaffenrot, V., Koopmans, P.J., 2022. Magnetization Transfer weighted laminar fMRI with multi-echo FLASH. *NeuroImage* 119725. <https://doi.org/10.1016/j.neuroimage.2022.119725>

Schulz, J., Fazal, Z., Metere, R., Marques, J.P., Norris, D.G., 2020. Arterial blood contrast (ABC) enabled by magnetization transfer (MT): a novel MRI technique for enhancing the measurement of brain activation changes. *bioRxiv*. <https://doi.org/10.1101/2020.05.20.106666>

Based on this literature, it seems that the proposed method will make the vein problem worse, not better. The authors could make it clearer how they reason that making GE-BOLD signals more extra-vascular weighted should help to reduce large vein effects.

The empirical evidence for the claim that flow crushing helps with the localization specificity should be made clearer. The response magnitude with and without flow crushing looks pretty much identical to me (see Fig, 6d). It's unclear to me what to look for in Fig. 5. I cannot discern any layer patterns in these maps. It's too noisy. The two maps of TE=43ms look like identical copies from each other. Maybe an editorial error?

The authors discuss bipolar crushing with respect to SE-BOLD where it has been previously applied. For SE-BOLD at UHF, a substantial portion of the vein signal comes from the intravascular compartment. So I agree that for SE-BOLD, it makes sense to crush the intravascular signal. For GE-BOLD however, this reasoning does not hold. For GE-BOLD (even at 3T), most of the vein signal comes from extravascular dephasing around large unspecific veins, and the bipolar crushing is not expected to help with this.

The authors would like to clarify that the velocity-nulling gradient is NOT designed to suppress all the contributions from intravascular blood. Instead, we tried to find a balance so that the VN gradient maximally suppressed the macrovascular signal in unspecific veins but minimally attenuated the microvascular signal in specific capillary bed. We acknowledge the reviewer's concern regarding the potential extravascular contributions from large, non-specific vessels. This aspect will be thoroughly evaluated and addressed in the revised manuscript. Additionally, we will make clarifications in other parts that may have cause the reviewer's misunderstandings.

1. The bipolar crushing is limited to one single direction of flow. This introduces a lot of artificial variance across the cortical folding pattern. This is not mentioned in the manuscript. There is an entire family of papers that perform layer-fMRI with black-blood imaging that solves this with a 3D contrast preparation (VAPER) that is applied across a longer time period, thus killing the blood signal while it flows across all directions of the vascular tree. Here, the signal crushing is happening with a 2D readout as a "snap-shot" crushing. This does not allow the blood to flow in multiple directions. VAPER also accounts for BOLD contaminations of larger draining veins by means of a tag-control sampling. The proposed approach here does not account for this contamination.

Chai, Y., Li, L., Huber, L., Poser, B.A., Bandettini, P.A., 2020. Integrated VASO and perfusion contrast: A new tool for laminar functional MRI. *NeuroImage* 207, 116358. <https://doi.org/10.1016/j.neuroimage.2019.116358>

Chai, Y., Liu, T.T., Marrett, S., Li, L., Khojandi, A., Handwerker, D.A., Alink, A., Muckli, L., Bandettini, P.A., 2021. Topographical and laminar distribution of audiovisual processing within human planum temporale. *Progress in Neurobiology* 102121. <https://doi.org/10.1016/j.pneurobio.2021.102121>

If I would recommend anyone to perform layer-fMRI with blood crushing, it seems that VAPER is the superior approach. The authors could make it clearer why users might want to use the unidirectional crushing instead.

We acknowledge that the degree of velocity nulling varies across the cortical folding pattern. We intend to discuss potential solutions to address this variance, and these may be implemented in the revised manuscript as appropriate. Furthermore, we will provide a comprehensive discussion on the advantages and disadvantages of both CBV-based and BOLD-based approaches.

1. The comparison with VASO is misleading. The authors claim that previous VASO approaches were limited by TRs of 8.2s. The authors might be advised to check the latest literature of the last years. Koiso et al. performed whole brain layer-fMRI VASO at 0.8mm at 3.9 seconds (with reliable activation), 2.7 seconds (with unconvincing activation pattern, though), and 2.3 (without activation). Also, whole brain layer-fMRI BOLD at 0.5mm and 0.7mm has been previously performed by the Juelich group at TRs of 3.5s (their TR definition is 'fishy' though).

Koiso, K., Müller, A.K., Akamatsu, K., Dresbach, S., Gulban, O.F., Goebel, R., Miyawaki, Y., Poser, B.A., Huber, L., 2023. Acquisition and processing methods of whole-brain layer-fMRI VASO and BOLD: The Kenshu dataset. *Aperture Neuro* 34. <https://doi.org/10.1101/2022.08.19.504502>

Yun, S.D., Pais-Roldán, P., Palomero-Gallagher, N., Shah, N.J., 2022. Mapping of whole-cerebrum resting-state networks using ultra-high resolution acquisition protocols. *Human Brain Mapping*. <https://doi.org/10.1002/hbm.25855>

Pais-Roldan, P., Yun, S.D., Palomero-Gallagher, N., Shah, N.J., 2023. Cortical depth-dependent human fMRI of resting-state networks using EPIK. *Front. Neurosci.* 17, 1151544. <https://doi.org/10.3389/fnins.2023.1151544>

The authors are correct that VASO is not advised as a turn-key method for lower brain areas, incl. Hippocampus and subcortex. However, the authors use this word of caution that is intended for inexperienced "users" as a statement that this cannot be performed. This statement is taken out of context. This statement is not from the academic literature.

It's advice for the 40+ user base that wants to perform layer-fMRI as a plug-and-play routine tool in neuroscience usage. In fact, sub-millimeter VASO is routinely being performed by MRI-physicists across all brain areas (including deep brain structures, hippocampus etc). E.g. see Koiso et al. and an overview lecture from a layer-fMRI workshop that I had recently attended: <https://youtu.be/kzh-nWXd54s?si=hoIjJLLixFUJ4g20&t=2401>

Thus, the authors could embed this phrasing into the context of their own method that they are proposing in the manuscript. E.g. the authors could state whether they think that their sequence has the potential to be disseminated across sites, considering that it requires slow offline reconstruction in Matlab? Do the authors think that the results shown in Fig. 6c are suggesting turn-key acquisition of a routine mapping tool? In my humble opinion, it looks like random noise, with most of the activation outside the ROI (in white matter).

Those literatures will be included and discussed in the revised manuscript. Furthermore, we are considering the exclusion of the LGN results presented in Figure 6, as they may divert attention from the primary focus of the study.

We are enthusiastic about sharing our imaging sequence, provided its usefulness is conclusively established. However, it's important to note that without an online reconstruction capability, such as the ICE, the practical utility of the sequence may be limited. Unfortunately, we currently don't have the manpower to implement the online reconstruction. Nevertheless, we are more than willing to share the offline reconstruction codes upon request.

1. The repeatability of the results is questionable. The authors perform experiments about the robustness of the method (line 620). The corresponding results are not suggesting any robustness to me. In fact, the layer profiles in Fig. 4c vs. Fig 4d are completely opposite. The location of peaks turns into locations of dips and vice versa. The methods are not described in enough detail to reproduce these results. The authors mention that their image reconstruction is done "using in-house MATLAB code" (line 634). They do not post a link to github, nor do they say if they share this code.

It is not trivial to get good phase data for fMRI. The authors do not mention how they perform the respective coil-combination. No data are shared for reproduction of the analysis.

There may have been a misunderstanding regarding the presentation in Figure 4, which illustrates the impact of TEs and the VN gradient. To enhance clarity and avoid further confusion, we will redesign this figure for improved comprehension.

The authors are open to sharing the MATLAB codes associated with our study. However, we were limited by manpower for refining and enhancing the readability of these codes for broader use.

Regarding the coil combination, we utilized an adaptive coil combination approach as described in the paper by Walsh DO, Gmitro AF, and Marcellin MW, titled 'Adaptive reconstruction of phased array MR imagery' (Magnetic Resonance in Medicine 2000; 43:682-690). The MATLAB code for this method was implemented by Dr. Diego Hernando. We will include a link for downloading this code in the revised manuscript for the convenience of interested readers.

1. The application of NORDIC is not validated. Previous applications of NORDIC at 3T layer-fMRI have resulted in mixed success. When not adjusted for the right SNR regime it can result in artifactual reductions of beta scores, depending on the SNR across layers. The authors could validate their application of NORDIC and confirm that the average layer-profiles are unaffected by the application of NORDIC. Also, the NORDIC version should be explicitly mentioned in the manuscript.

Akbari, A., Gati, J.S., Zeman, P., Liem, B., Menon, R.S., 2023. Layer Dependence of Monocular and Binocular Responses in Human Ocular Dominance Columns at 7T using VASO and BOLD (preprint). *Neuroscience*. <https://doi.org/10.1101/2023.04.06.535924>

Knudsen, L., Guo, F., Huang, J., Blicher, J.U., Lund, T.E., Zhou, Y., Zhang, P., Yang, Y., 2023. The laminar pattern of proprioceptive activation in human primary motor cortex. *bioRxiv*. <https://doi.org/10.1101/2023.10.29.564658>

During our internal testing, we observed that the NORDIC denoising process did not alter the activation patterns. These findings will be incorporated into the revised manuscript. The details of NORDIC will be provided as well.

Reviewer #2 (Public Review):

[...] The well-known double peak feature in M1 during finger tapping was used as a test-bed to evaluate the spatial specificity. They were indeed able to demonstrate two distinct peaks in group-level laminar profiles extracted from M1 during finger tapping, which was largely free from superficial bias. This is rather intriguing as, even at 7T, clear peaks are usually only seen with spatially specific non-BOLD sequences. This is in line with their simple simulations, which nicely illustrated that, in theory, intravascular macrovascular signals should be suppressible with only minimal suppression of microvasculature when small b-values of the VN gradients are employed. However, the authors do not state how ROIs were defined making the validity of this finding unclear; were they defined from independent criteria or were they selected based on the region mostly expressing the double peak, which would clearly be circular? In any case, results are based on a very small sub-region of M1 in a single slice - it would be useful to see the generalizability of superficial-bias-free BOLD responses across a larger portion of M1.

Given the individual variations in the location of the M1 region, we opted for manual selection of the ROI. In the revised manuscript, we plan to explore and implement an independent criterion for ROI selection to enhance the objectivity and reproducibility of our methodology.

As repeatedly mentioned by the authors, a laminar fMRI setup must demonstrate adequate functional sensitivity to detect (in this case) BOLD responses. The sensitivity evaluation is unfortunately quite weak. It is mainly based on the argument that significant activation was found in a challenging sub-cortical region (LGN). However, it was a single participant, the activation map was not very convincing, and the demonstration of significant activation after considerable voxel-averaging is inadequate evidence to claim sufficient BOLD sensitivity. How well sensitivity is retained in the presence of VN gradients, high acceleration factors, etc., is therefore unclear. The ability of the setup to obtain meaningful functional connectivity results is reassuring, yet, more elaborate comparison with e.g., the conventional BOLD setup (no VN gradients) is warranted, for example by comparison of tSNR, quantification and comparison of CNR, illustration of unmasked-full-slice activation maps to compare noise-levels, comparison of the across-trial variance in each subject, etc. Furthermore, as NORDIC appears to be a cornerstone to enable submillimeter resolution in this setup at 3T, it is critical to evaluate

its impact on the data through comparison with non-denoised data, which is currently lacking.

We appreciate the reviewer's comments. Those issues will be addressed carefully.

Reviewer #3 (Public Review):

[...] Weaknesses:

- *Although the VASO acquisition is discussed in the introduction section, the VN-sequence seems closer to diffusion-weighted functional MRI. The authors should make it more clear to the reader what the differences are, and how results are expected to differ. Generally, it is not so clear why the introduction is so focused on the VASO acquisition (which, curiously, lacks a reference to Lu et al 2013). There are many more alternatives to BOLD-weighted imaging for fMRI. CBF-weighted ASL and GRASE have been around for a while, ABC and double-SE have been proposed more recently.*

The principal distinction between DW-fMRI and our methodology lies in the level of the b-value employed. DW-fMRI typically measures cellular swelling by utilizing a b-value greater than 1000 s/mm² (e.g. 1800). Conversely, our Velocity Nulling functional MRI (VN-fMRI) approach continues to assess hemodynamic responses, utilizing a smaller b-value specifically for the suppression of signals from draining veins. In addition, other layer-fMRI methods will be discussed.

- *The comparison in Figure 2 for different b-values shows % signal changes. However, as the baseline signal changes dramatically with added diffusion weighting, this is rather uninformative. A plot of t-values against cortical depth would be much more insightful.*
- *Surprisingly, the %-signal change for a b-value of 0 is not significantly different from 0 in the gray matter. This raises some doubts about the task or ROI definition. A finger-tapping task should reliably engage the primary motor cortex, even at 3T, and even in a single participant.*
- *The BOLD weighted images in Figure 3 show a very clear double-peak pattern. This contradicts the results in Figure 2 and is unexpected given the existing literature on BOLD responses as a function of cortical depth.*

In our study, the TE in Figure 2 is shorter than that in Figure 3 (33 ms versus 43 ms). It has been reported in the literature that BOLD fMRI with a shorter TE tends to include a greater intravascular contribution. Acknowledging this, we plan to repeat the experiments with a controlled TE to ensure consistency in our results.

- *Given that data from Figures 2, 3, and 4 are derived from a single participant each, order and attention affects might have dramatically affected the observed patterns. Especially for Figure 4, neither BOLD nor VN profiles are really different from 0, and without statistical values or inter-subject averaging, these cannot be used to draw conclusions from.*

The order of the experiments were randomized to ensure unbiased results.

It is important to note that the error bars presented in Figures 2, 3, and 4 do not represent the standard deviation of the residual fitting error. Instead, they illustrate the variation across voxels within a specific layer. This approach may lead to the error bars being influenced by

the selection of the Region of Interest (ROI). In light of this, we intend to refine our statistical methodologies in the revised manuscript to address this issue.

- *In Figure 5, a phase regression is added to the data presented in Figure 4. However, for a phase regression to work, there has to be a (macrovascular) response to start with. As none of the responses in Figure 4 are significant for the single participant dataset, phase regression should probably not have been undertaken. In this case, the functional 'responses' appear to increase with phase regression, which is contra-intuitive and deserves an explanation.*
- *Consistency of responses is indeed expected to increase by a removal of the more variable vascular component. However, the microvascular component is always expected to be smaller than the combination of microvascular + macrovascular responses. Note that the use of %signal changes may obscure this effect somewhat because of the modified baseline. Another expected feature of BOLD profiles containing both micro- and macrovasculature is the draining towards the cortical surface. In the profiles shown in Figure 7, this is completely absent. In the group data, no significant responses to the task are shown anywhere in the cortical ribbon.*
- *Although I'd like to applaud the authors for their ambition with the connectivity analysis, I feel that acquisitions that are so SNR starved as to fail to show a significant response to a motor task should not be used for brain wide directed connectivity analysis.*

We agree that exploring brain-wide directed functional connectivity may be overly ambitious at this stage, particularly before the VN-fMRI technique has been comprehensively evaluated and validated. In the revised manuscript, we will focus more on examining the characteristics of the layer-dependent BOLD signal rather than delving into layer-dependent functional connectivity.